

Security Model of Authentication at the Physical Layer and Performance Analysis over Fading Channels

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Abstract—Compared with conventional authentication schemes, which utilize cryptographic mechanisms and operate at an upper layer, authentication at the physical (PHY) layer possesses many advantages, such as, higher information-theoretic security by introducing uncertainty to an adversary, better efficiency, and better compatibility by avoiding any operations at the upper layer. An effective PHY-layer authentication scheme should consider covertness, security, and robustness together. However, in published literature, these three properties have been separately analyzed. This paper proposes a generic security model for PHY-layer authentication, and we design a new systematic metric, which is referred to as the probability of security authentication (PSA). According to the proposed generic security model, we can not only systematically analyze the effect of the parameter of a certain PHY-layer authentication scheme on the final performance, but also fairly compare the performance of different PHY-layer authentication schemes under the same channel conditions at the legitimate and adversary receivers. For performance analysis, the probability of detection (PD) and the probability of false alarm (PFA) of two PHY-layer authentication schemes over fading channels are defined, and their closed-form expressions are well derived. On the basis of the proposed generic security model, a systematic performance comparison between different PHY-layer schemes is provided.

Index Terms—Security model, PHY-layer authentication, probability of security authentication, fading channel

1 INTRODUCTION

IN modern wireless systems, the ability to verify the authenticity of a transmitter is a fundamental security requirement. It is especially important in wireless systems, because the open nature of a shared medium presents more security vulnerabilities for an adversary to eavesdrop, jam, and impersonate. Recently, increasing attention has been paid to the authentication at the physical (PHY) layer, owing to its two major advantages over conventional authentication technology based on the cryptographic tool at the upper-layer. First, the assumption of a computational limitation for cryptographic tools has been gradually weakened with the advancement in computational power and cryptanalysis algorithms. For example, the authentication tag based on the cryptographic hash function MD5 can be forged in a few seconds using an off-the-shelf personal computer [1]. The reason behind this is that the attack on cryptographic tools relies on noiseless access to both the data and authentication tag, which allows an adversary to search information regarding the secret key. However, authentication at the PHY-layer protects the authentication tag by presenting the adversary with the

noisy observation only, which introduces an uncertainty regarding the secret key that cannot be broken by any increase in the computational ability. Second, the authentication at the PHY-layer enables a legitimate receiver to quickly distinguish between a legitimate transmitter and a rogue one without complete upper layer processing, which requires many extra network resources. Thus, the authentication at the PHY-layer is especially useful in heterogeneous coexistence environments, e.g., 5G systems, where incompatible systems may not be able to decode the higher layer signal of each other [2], [3].

An effective authentication scheme should consider covertness, security, and robustness together. With regard to covertness, an unaware receiver must not be able to detect that the authenticated signal is anomalous. The need for the covert property arises when the authentication is piggybacked onto an existing system. For example, some unaware receivers do not know the authentication scheme and cannot authenticate the transmitter at the PHY-layer, but he or she should be able to demodulate and decode the message signal that can be authenticated at the upper layers [4], [5]. Moreover, covert authentication at the PHY-layer may be used together with other security techniques at the upper layer to provide a more secure system. Regarding security, an adversary must not be able to easily break the authentication through different attacks, including jamming, substitution, and impersonation attacks. With regard to robustness, because transmissions are performed in random fading environments, we highly desire that the authentication scheme is resistant to channel fading and noise

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effects. An authentication scheme that can operate in the midst of interference is robust.

In the literature, the authentication schemes designed at the PHY-layer can be broadly divided into two categories: passive and active styles. In the passive style, the intrinsic features of the communication system are exploited to authenticate the transmitter as a unique signature, e.g., RF signal characteristics were utilized in [6], [7], [8], [9], and channel characteristics were utilized in [10], [11], [12], [13], [14], [15], [16]. In [17], the reliability and differentiability of these intrinsic features were evaluated using a comprehensive theoretical model and through some in-laboratory experiments. The schemes in the first category have been verified in well-designed laboratory environments, however they are sensitive to external variables, such as temperature, channel variation and possible malicious attacks, which limit their application in practice. This paper focuses on the second category.

In the active style, an artificial tag is embedded in the message signal and is then extracted at the receiver. An authentication tag was appended to the data in [18] using a time-division multiplexing method, referred to as the Auth-TDM scheme. On the one hand, the Auth-TDM scheme requires additional transmission time; On the other hand, in the Auth-TDM scheme, the authentication tag is easily exposed to an adversary due to the same signal-to-noise ratio (SNR) as that of the message signal. In [19], for an orthogonal frequency-division multiplexing (OFDM) system, a cyclo-stationary signature was generated by repeating some message symbols over the sub-carriers according to the authentication tag, which obviously wasted the message throughput. In [20], the frequency offset was modified according to the authentication tag, but the rate of authentication tag transmitted per second was very low. In [5], for a pre-coded duobinary signaling system, some initialization bits were modified according to the authentication tag. This scheme makes recovery of the message signal by an unaware receiver difficult, if knowledge of this authentication scheme is unavailable, which obviously violates the requirement of covertness. Recently, the widely applied scheme in the active style has been the authentication scheme with a superimposed tag, referred to as the Auth-SUP scheme, which concurrently transmits an additional authentication tag with the messages [4], [21]. The experimental results of the Auth-SUP scheme have been implemented and analyzed using the platform of the software-defined radios (SDR) in [22], [23]. By making a tradeoff among covertness, security, and robustness, the Auth-SUP scheme can satisfy the requirement of the effective authentication scheme described above.

It should be noted that in the literature, the covertness, security, and robustness of a PHY-layer authentication are often separately analyzed. Thus, a systematic analysis on the effect of the parameters of a PHY-layer authentication scheme on the final performance becomes difficult. Moreover, a fair comparison on the performance of different PHY-layer authentication schemes under the same channel conditions is also difficult. The motivation of this paper is to fill this gap in this regard.

The main contributions of this paper are summarized as follows:

- 1) A new generic security model of PHY-layer authentication by considering covertness, security and robustness together is proposed. On the basis of the proposed generic security model, a new systematic metric is also designed, which is referred to as the secure probability of authentication accuracy (PSA). Note that, based on the proposed generic security model, we not only quantitatively and systematically analyze the effect of the parameter of a certain PHY-layer authentication scheme on the final performance, but also fairly compare the performance of different PHY-layer authentication schemes under the same channel conditions at the legitimate and adversary receivers.
- 2) For performance analysis, the probability of detection (PD) and probability of false alarm (PFA) of two PHY-layer authentication schemes over fading channels have been defined, and their closed-form expressions have been well derived. On the basis of the proposed generic security model, a systematic performance comparison between two schemes is provided.

The remainder of this paper is organized as follows. Section 2 introduces the system model and challenges. The new generic security model for PHY-layer authentication is proposed and a new metric of a secure PHY-layer authentication scheme is then designed and presented in Section 3. Section 4 presents the analysis of the Auth-SUP scheme based on the proposed generic security model. In Section 5, an modified version of the Auth-TDM scheme is given for comparison purposes, which is referred to as the Auth-REP scheme and is analyzed based on the proposed generic security model. The performance comparison between the Auth-SUP and Auth-REP schemes based on the proposed generic security model is provided and some interesting observations are presented in Section 6. Concluding remarks are provided in Section 7.

A summary of the variables and notations used in this paper is given in Table 1.

2 SYSTEM MODEL

2.1 System Model

We assume that the PHY-layer authentication scenario considered in this paper is shown in Fig. 1, where four nodes share a wireless medium. Alice sends messages to Bob; Carol and Eve listen. This scenario has no privacy. Hence, Carol and Eve can understand what Alice is sending to Bob. We assume that Alice and Bob agree on an authentication scheme with a shared secret key that allows Bob to verify the messages from Alice. To authenticate, Alice sends a proof of authentication, called a tag, together with each message for Bob's verification. The tags reflect knowledge of the secret key shared between Alice and Bob. The authenticated signal is called a tagged signal, whereas the transmitted signal without tag is called a normal signal. Carol is an unaware receiver because she does not know the authentication scheme. Carol cannot authenticate the tagged signal; however, she can recover the data messages. We assume that Eve is an adversary that is aware of the authentication scheme; however, she cannot authenticate the tagged signal because she does not know the secret key.

TABLE 1
List of Variables and Notations

Variable /Notation	Meaning	Variable /Notation	Meaning
L	Length of block	$\mathbf{b}$	Message
$\mathbf{c}$	Encoded signal	$\mathbf{s}$	Modulated signal
$\mathbf{x}$	Transmitted signal	$\mathbf{t}$	Tag
$\mathbf{y}$	Received signal	$\mathbf{k}$	Secret key
$\mathbf{n}$	Receiver noise	$\mathbf{r}$	Residual signal
ρ_s^2, ρ_t^2	Parameters of the Auth-SUP scheme	L_R	Parameters of the Auth-REP scheme
h	Channel response	ε_{FA}	Bound of PFA
$\mathbb{E}\{\cdot\}$	Expectation operator	$\text{var}\{\cdot\}$	Variance operator
$\mathbb{H}(\cdot)$	Binary entropy operator	$\text{Pr}(\cdot)$	Probability measure
$Q(\cdot)$	Tail distribution function of a standard normal distribution	$\mathbb{R}\{\cdot\}$	Real value extractor
$(\cdot)^H$	Hermitian transpose operator	$(\cdot)^*$	Conjugate operator

Eve wishes to disrupt the authentication process by causing Bob to either reject authentic messages or accept unauthentic messages.

A flat block-fading channel is assumed such that different message blocks experience independent fades. The source messages are blocks of L symbols denoted by $\mathbf{b}_i$ for the i th block, which are independent and identically distributed (i.i.d.) random variables (RVs). Through channel coding, $\mathbf{b}_i$ is encoded as $\mathbf{c}_i$. Then, through modulation, $\mathbf{c}_i$ is modulated as $\mathbf{s}_i$. For a normal signal, the transmitted signal $\mathbf{x}_i$ can be represented as $\mathbf{x}_i = \mathbf{s}_i$. Meanwhile, a tagged signal can be expressed as

$$\mathbf{x}_i = \text{Emb}(\mathbf{s}_i, \mathbf{t}_i, \phi_i), \quad (1)$$

where embedding function $\text{Emb}(\cdot)$ is determined by the specific PHY-layer authentication scheme with parameter ϕ_i . For simplicity, we assume that $\mathbb{E}\{|\mathbf{x}_i|^2\} = \mathbb{E}\{|\mathbf{s}_i|^2\} = L$. The tag signal $\mathbf{t}_i$ is generated by a hash function $g(\cdot)$ with secret key $\mathbf{k}$ shared between Alice and Bob. In [4] and [21], the tag generation is expressed as

$$\mathbf{t}_i = g(\mathbf{s}_i, \mathbf{k}). \quad (2)$$

Through a flat block-fading channel, the receiver observes the transmitted signal as

$$\mathbf{y}_i = h_i \mathbf{x}_i + \mathbf{n}_i, \quad (3)$$

where h_i is the channel response, which is a zero-mean complex Gaussian RV with variance σ_h^2 . We denote $\sigma_h^2 \propto d^{-\alpha_d}$, where $\alpha_d \geq 2$ is the channel pathloss exponent, and d is the distance between the transmitter and receiver. $\mathbf{n}_i$ is a zero-mean complex white Gaussian noise with variance σ_n^2 . The channel SNR for the i th block is denoted as $\gamma_i = |h_i|^2 / \sigma_n^2$. After estimating the channel, the receiver uses its channel estimation $\hat{h}_i$ to estimate the i th transmitted signal as

$$\hat{\mathbf{x}}_i = \frac{\hat{h}_i^*}{|\hat{h}_i|^2} \mathbf{y}_i. \quad (4)$$

Here, to simplify the analysis, we assume that the receiver has a perfect channel estimation, i.e., $\hat{h}_i = h_i$, because this assumption is reasonable in a block fading channel, which

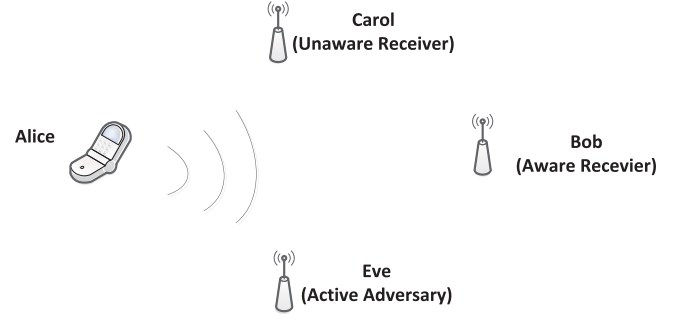


Fig. 1. PHY-layer authentication scenario considered in this paper, where four nodes share a wireless medium. Alice sends messages to Bob, while Carol and Eve listen.

is similar to that [4]. Then, by sequential demodulation and decoding, the receiver can obtain demodulated output $\hat{\mathbf{c}}_i$ and decoded output $\hat{\mathbf{b}}_i$.

For Bob to authenticate the transmitter, the receiver should extract residual signal $\mathbf{r}_i$ from $\mathbf{y}_i$, which is expressed as

$$\mathbf{r}_i = \text{Ext}(\mathbf{y}_i, \mathbf{k}, \phi_i), \quad (5)$$

where extracting function $\text{Ext}(\cdot)$ is also determined by the specific PHY-layer authentication scheme with parameter ϕ_i and secret key $\mathbf{k}$. Then Bob performs a threshold test using the following hypotheses

$$\begin{aligned} H_0 : & \mathbf{t}_i \text{ is not present in } \mathbf{r}_i \\ H_1 : & \mathbf{t}_i \text{ is present in } \mathbf{r}_i. \end{aligned} \quad (6)$$

Two error types are possible in the decision making. Accepting hypothesis H_1 when H_0 is true is called a false alarm; The PFA is denoted by $P_{FA, \text{Bob}}$. In contrast, accepting hypothesis H_0 when H_1 is true is called a missed detection; The probability of missed detection (PMD) is denoted by $P_{MD, \text{Bob}}$. The optimal decision rule in (6) is given by the Neyman-Pearson theorem, i.e., the optimal threshold is determined by maximizing the PD $P_{D, \text{Bob}}$, while making $P_{FA, \text{Bob}} \leq \varepsilon_{PFA, \text{Bob}}$, where $\varepsilon_{PFA, \text{Bob}}$ is the upper bound of the PFA allowed by Bob. Mathematically, the optimization problem can be expressed as

$$\begin{aligned} & \max P_{D, \text{Bob}} \\ & \text{subject to } P_{FA, \text{Bob}} \leq \varepsilon_{PFA, \text{Bob}}. \end{aligned} \quad (7)$$

We suppose that Eve knows the authentication scheme that Alice and Bob are using; however, she does not know the secret key. Before she starts an attack, e.g., substitution and impersonation attacks, she should analyze her observations to estimate the secret key. To highlight the advantage of the PHY-layer authentication, we assume that Eve has unlimited computational resources. Thus, she can estimate the secret key, if she can observe the tagged signal without any error.

2.2 Challenges

An effective PHY-layer authentication scheme must consider three properties: covertness, security and robustness, as mentioned in the Introduction section. In a practical

scenario, achieving optimal performance for all three properties is challenging. In general, the first two properties have different requirements on the authentication parameter at the PHY-layer compared with the third property. For example, in the Auth-SUP scheme [4], covertness and security require low tag energy, whereas robustness requires the opposite. The reason behind this is because by allowing more errors in the authentication process, Eve has a better opportunity to sneak in her own messages.

However, in published literature, these three properties have been separately analyzed. Thus, difficulty arises in not only the systematic analysis of the effect of the parameter of certain PHY-layer authentication scheme on the final performance, but also the fair comparison of the performance of various PHY-layer authentication schemes based on a uniform standard over the three properties. This paper attempts to fill the gap in this regard by proposing a generic security model for PHY-layer authentication.

3 SECURITY MODEL OF PHY-LAYER AUTHENTICATION

This paper proposes a generic security model for a PHY-layer authentication scheme by considering the three above-mentioned properties together, which includes two steps. In the first step, we simultaneously analyze the authentication accuracy at Bob, η_{Bob} , by considering robustness, and the measure of feasible authentication information leakage (FAIL) at Eve η_{Eve} by considering covertness and security. In the second step, on the basis of the two measures, we design a new metric, referred to as probability of security authentication (PSA), namely P_{SA} , which is explicitly defined in the following definition.

Definition 1. For a certain PHY-layer authentication scheme, if two measures η_{Bob} and η_{Eve} can be analyzed, the value of P_{SA} is calculated as

$$P_{\text{SA}} = \max\{\eta_{\text{Bob}} - \eta_{\text{Eve}}, 0\}. \quad (8)$$

The idea behind Definition 1 is the actual utilization of the method to establish the secrecy capacity in the PHY-layer security defined in [24]. The PSA denotes the difference between the reliable authentication accuracy between Alice and Bob from which we subtract the information leakage to Eve. A secure PHY-layer authentication should seek for high PSA, and it can actually be viewed as a counterpart of the traditional PHY-layer authentication, when robustness, covertness, and security are imposed.

In this paper, authentication accuracy η_{Bob} represents the robustness of a certain PHY-layer authentication scheme, which is simply denoted as the PD at Bob $P_{\text{D,Bob}}$ with side information. However, the definition of FAIL measure η_{Eve} is more complicated, since it should represent the covertness and security. Although Eve does not initially have secret key $\mathbf{k}$ shared by Alice and Bob, she gains key information by observing their communications. When she learns $\mathbf{k}$, she is able to impersonate Alice at will by generating legitimate tagged signal for her messages using (1). Therefore, we are interested on how hard it is for Eve to recover the key information. Moreover, we further want that the proposed FAIL

measure considers the feasibility of Eve's attack by taking account of her computational limits.

In this paper, the FAIL measure η_{Eve} is established by considering two factors. First, we use the key equivocation to measure how much is Eve's uncertainty about the key $\mathbf{k}$ given her observations, assuming that her computational resources are unlimited. Second, Eve should have the capability to distinguish a normal signal from a tagged one, since Alice acting as Eve's opponent does not always unwisely embed the authentication tag into all message blocks. In other words, although we assume that Eve has unlimited computational resources when we calculate the key equivocation, it is impractical and inefficient. Thus, a wise Eve can launch a feasible attack with limited computational resources. Moreover, the second factor also represents the covertness of a PHY-layer authentication scheme, since as the covertness of a PHY-layer authentication scheme enhances, the value of the second factor decreases. In summary, the first factor represents the authentication information leakage, whereas the second factor represents the feasibility of attacks and its covertness. The details of the two factors as well as the PSA are described in the following sections.

We note that, more importantly, although it is a challenging task by using single metric to describe a full enumeration of all attacks launched by Eve, the proposed security model deals with most part of main attacks in PHY-layer authentication, such as jamming attacks, substitution attacks and impersonation attacks. Specifically, for a jamming attack, Eve can launch it without learning the secret shared between Alice and Bob, e.g., jamming attacks, its effect can be reflected by the first measure of the proposed security model since it deteriorates the SNR at Bob. For a substitution attack or impersonation attack, Eve should first learn the secret shared between Alice and Bob, their effect can be reflected by the second measure of the proposed security model.

3.1 Key Equivocation

For a given data frame, the number of distinct authentication tags that can be assigned to it is bounded by the number of keys. If sufficiently many tags are observed without noise, as in the case with conventional cryptographic authentication scheme, Eve will eventually obtain the key $\mathbf{k}$ and can mount successful impersonation or substitution attacks [25]. However, if the tags are viewed as a noise and, therefore, with error (as existing in the PHY-layer authentication schemes), the effort required to recover the secret key can be significantly increased. The following analysis quantifies this increase.

We assume that Eve knows the tag generating hash function $g(\cdot)$, and she is able to recover the message s_i without error due to the covertness requirement, just as Bob can. The key equivocation calculated by conditional entropy [26] is a method that measures how much Eve's uncertainty about the key decreases given her observations, assuming her computational resources are unlimited. For example, given an M -bit key drawn from uniform distribution, if Eve has zero key equivocation, she has no uncertainty about the key and therefore exactly knows it. On the other hand, if Eve has M bits of key equivocation, she knows nothing about the key.

First let us consider the key equivocation under noiseless observations of both s_i and t_i

$$\mathbb{H}(\mathbf{k}|\mathbf{s}_i, \mathbf{t}_i) = \sum_{s \in \mathcal{S}, t \in \mathcal{T}} \Pr(\mathbf{s}_i, \mathbf{t}_i) \mathbb{H}(\mathbf{k}|s, t), \quad (9)$$

where $\mathbb{H}(\mathbf{k}|s, t)$ is calculated as

$$\mathbb{H}(\mathbf{k}|s, t) = - \sum_{k \in \mathcal{K}} \Pr(k|s, t) \log \Pr(k|s, t), \quad (10)$$

where $\Pr(k|s, t) \triangleq \Pr(\mathbf{k} = k | \mathbf{s}_i = s, \mathbf{t}_i = t)$. $\mathcal{S}$, $\mathcal{T}$, and $\mathcal{K}$ are the domains of $\mathbf{s}_i$, $\mathbf{t}_i$, and $\mathbf{k}$, respectively. From (10), it follows that, for a particular choice of s and t , positive key equivocation arises when multiple keys map s to the same t , i.e., when $g(\cdot)$ is not invertible in k . For this condition to be true, we must have $|\mathcal{K}| > |\mathcal{T}|$ which is also the requirement of cryptanalysis algorithms at upper layer, where $|\cdot|$ represents the set cardinality. Zero key equivocation is the worst case for security because brute-force attacks are guaranteed to succeed.

To highlight the advantage of PHY-layer authentication, we assume that $|\mathcal{K}| \leq |\mathcal{T}|$, which makes $\mathbb{H}(\mathbf{k}|s, t) = 0$ due to Eve's unlimited computational resources. We now consider how noisy observations of the tag can increase Eve's key equivocation. Because we assume that Eve knows the transmitted message s and the set of possible keys $\mathcal{K}$, she can step through each possible key to enumerate the possible tags corresponding to, denoted as

$$t_i = g(s, k_i), \forall k_i \in \mathcal{K}. \quad (11)$$

Because $g(\cdot)$ is a one-way function, the distribution of $\{t_i\}$ is approximately uniform over $\mathcal{T}$. Further, because we assume that $\mathbb{H}(\mathbf{k}|s, t) = 0$, $\{t_i\}$ contains no duplicates and $|\{t_i\}| = |\mathcal{K}|$.

Suppose that Eve makes an estimate of the tag $\hat{t}$ from her observation $\mathbf{y}_i$. She calculates the Hamming distance $h(\hat{t}, t_i)$ from each valid tag t_i . An i.i.d. tag bit-error probability can be calculated as $p_e = Q(\sqrt{\gamma_i})$, where $Q(\cdot)$ is the tail probability function of the standard normal distribution, and γ_i is the tag-to-noise ratio (e.g., $\gamma_i = \rho_i^2 \gamma_{i, \text{Bob}}$ for the Auth-SUP scheme). The probability that t_i is transmitted given that $\hat{t}$ is observed is given by the binomial probability

$$\Pr(t_i|\hat{t}) = p_e^{h(\hat{t}, t_i)} (1 - p_e)^{\log_2 |\mathcal{T}| - h(\hat{t}, t_i)}. \quad (12)$$

When $|\mathcal{T}|$ is strictly greater than $|\mathcal{K}|$, $\{t_i\} \subset \mathcal{T}$, $\Pr(t_i|\hat{t})$ is not a probability distribution because $\sum_{t_i \in \mathcal{T}} \Pr(t_i|\hat{t}) < 1$. Thus, we consider the scaled version to represent a valid probability distribution, which is expressed as

$$f(t_i|\hat{t}) = \frac{\Pr(t_i|\hat{t})}{\sum_{t_i \in \mathcal{T}} \Pr(t_i|\hat{t})}. \quad (13)$$

Since we assume that $\mathbb{H}(\mathbf{k}|s, t) = 0$, the distribution over the keys k_i is exactly the distribution $f(\cdot)$ over the corresponding tags t_i . Thus, the key equivocation given the observation is denoted as

$$\begin{aligned} \mathbb{H}(\mathbf{k}|\mathbf{y}_i) &\cong \mathbb{H}(\mathbf{k}|\mathbf{s}_i, \hat{\mathbf{t}}_i) = - \sum_{k_i \in \mathcal{K}} f(k_i|\hat{t}) \log_2 f(k_i|\hat{t}) \\ &= - \sum_{k_i \in \mathcal{K}} f(t_i|\hat{t}) \log_2 f(t_i|\hat{t}). \end{aligned} \quad (14)$$

3.2 Eve's Distinguishing Capability

Before Eve starts an attack, e.g., substitution and impersonation attacks, she should analyze her observations to estimate the secret key. However, Alice acting as a wise designer can wisely embed the authentication tag into a random portion of message blocks, to lower Eve's estimation accuracy. Thus, Eve acting as Alice's opponent should first distinguish whether a tagged signal is coming or not to avoid extravagant computational resources. Here, Eve also performs a threshold test using the following hypotheses to represent her distinguishing capability

$$\begin{aligned} H_0 : & \mathbf{y}_i \text{ is a normal signal} \\ H_1 : & \mathbf{y}_i \text{ is a tagged signal.} \end{aligned} \quad (15)$$

Two error types are also possible here in the decision making, which is similar to that in (6). The difference between the two threshold tests in (6) and (15) is that the optimal decision rule of the former is determined using the secret side information shared by Alice and Bob, e.g., ϕ_i and $\mathbf{k}$, whereas that of the latter is determined without such side information.

We note that, in a reliable wireless communication system, a PHY-layer should provide error-free bit sequence to the upper-layer application, i.e., $\hat{\mathbf{b}}_i = \mathbf{b}_i$. Then by sequentially re-encoding and re-modulating to $\hat{\mathbf{b}}_i$, Eve also obtains $\mathbf{s}_i$ without any error for a normal signal. To satisfy the covertness requirement of the PHY-layer authentication, Eve should obtain $\mathbf{s}_i$ without error for a tagged signal. Thus, Eve constructs test statistic $\delta_{i, \text{Eve}}$ by match-filtering the residual using the estimated tag as

$$\delta_{i, \text{Eve}} = \mathbb{R}\{\tau_{i, \text{Eve}}\} = \mathbb{R}\{\mathbf{s}_i^H \mathbf{s}_i - \hat{\mathbf{x}}_i^H \mathbf{s}_i\}, \quad (16)$$

where $\mathbb{R}\{\cdot\}$ represents the operation of extracting the real part, and $\hat{\mathbf{x}}_i$ is defined in (4). Because any PHY-layer authentication scheme embeds the tag in the transmitted signal by slightly modifying some properties of the transmitted signal, the threshold test in (15) based on (16) can be regarded as a binary hypothesis problem with different mean but the same variance, which will be explained in the following sections. We note that the reasons of calculating test statistic $\delta_{i, \text{Eve}}$ using the correlation operation are twofold. First, the correlation between $\hat{\mathbf{x}}_i$ and $\mathbf{s}_i$ explicitly quantifies how much the space of the normal signal is occupied by the authentication tag for a PHY-layer authentication scheme with active style, i.e., the lower the correlation value is, the more the normal signal space is occupied. Second, the test statistic using the correlation operation provides tractable closed-form results for existing PHY-layer authentication schemes with active style. In the future, the test statistic may of course be re-designed for a certain advanced PHY-layer authentication scheme.

The decision of authenticity $\varphi_{i, \text{Eve}}$ for the i th block based on $\delta_{i, \text{Eve}}$ is made according to

$$\varphi_{i, \text{Eve}} = \begin{cases} 0, & \delta_{i, \text{Eve}} < \theta_{i, \text{Eve}} \\ 1, & \delta_{i, \text{Eve}} \geq \theta_{i, \text{Eve}}, \end{cases} \quad (17)$$

where $\theta_{i, \text{Eve}}$ is the test's threshold using the hypotheses defined in (15). According to the Neyman-Pearson principle again, the optimal detector imposes a bound on the PFA at Eve, namely $P_{\text{FA}, \text{Eve}} \leq \varepsilon_{\text{PFA}, \text{Eve}}$, while maximizing the PD at Eve, i.e., $P_{\text{D}, \text{Eve}}$. We note that the value of $P_{\text{D}, \text{Eve}}$ represents

the distinguishing capability, while the value of $P_{\text{FA},\text{Eve}}$ represents her limit of computational resources.

3.3 Probability of Security Authentication

Simultaneously, the PD at Bob, $P_{\text{D},\text{Bob}}$, can be obtained by solving the optimization problem defined in (6) and (7). In summary, we straightforwardly set the authentication accuracy at Bob as

$$\eta_{\text{Bob}} = P_{\text{D},\text{Bob}}, \quad (18)$$

whereas the FAIL measure is calculated as

$$\eta_{\text{Eve}} = P_{\text{D},\text{Eve}} I_{\text{Eve}}, \quad (19)$$

where

$$I_{\text{Eve}} = \left(1 - \frac{\mathbb{H}(\mathbf{k}|\mathbf{y}_i)}{\log_2 |\mathcal{T}|}\right), \quad (20)$$

is the ratio of authentication information leakage, and $\mathbb{H}(\mathbf{k}|\mathbf{y}_i)/\log_2 |\mathcal{T}|$ denotes the normalized key equivocation. The value of I_{Eve} falls into $[0, 1]$. When $\mathbb{H}(\mathbf{k}|\mathbf{y}_i) = 0$ is its achievable minimum value, we obtain $I_{\text{Eve}} = 1$, which indicates that no key equivocation is present and the authentication information is totally exposed to Eve. In contrast, when $\mathbb{H}(\mathbf{k}|\mathbf{y}_i) = \log_2 |\mathcal{T}|$ is its achievable maximum value, we obtain $I_{\text{Eve}} = 0$, which indicates that Eve gains no key information and no authentication information is available to be leaked to Eve. The first step of the proposed generic framework security model is then completed.

According to (18) and (19), the PSA defined in Definition 1 can be specified as

$$\begin{aligned} P_{\text{SA}} &= \max\{P_{\text{D},\text{Bob}} - P_{\text{D},\text{Eve}} I_{\text{Eve}}, 0\}. \\ \text{subject to } &P_{\text{FA},\text{Bob}} \leq \varepsilon_{\text{PFA},\text{Bob}} \\ &P_{\text{FA},\text{Eve}} \leq \varepsilon_{\text{PFA},\text{Eve}} \end{aligned} \quad (21)$$

The PSA expression in Definition 1 allows us to make several important observations. P_{SA} is non-zero that provided $\eta_{\text{Bob}} - \eta_{\text{Eve}}$ is positive; in particular, when the tag signal is easily detected by Eve with high possibility, then P_{SA} may approach zero. Hence, whereas the PHY-layer authentication does provide some level of information-theoretic secrecy, it requires Alice and Bob to have an advantage over Eve at the PHY-layer itself. In contrast, the authentication of the cryptographic mechanisms cannot provide information-theoretic secrecy. Based on (21), we can systematically analyze the effect of the parameter of a certain PHY-layer authentication scheme on the final performance rather than separately consider the covertness, security, and robustness. Moreover, we can fairly compare the performance of different PHY-layer authentication schemes under the same $\varepsilon_{\text{PFA},\text{Bob}}$ and $\varepsilon_{\text{PFA},\text{Eve}}$.

The major contribution of establishing a PSA expression is to facilitate a system designer to work out a more secure authentication scheme. We suppose that one suspicious user (e.g., Eve) is near Alice. Although Alice cannot make sure whether she is an adversary user or not, Alice can evaluate the potential threats from the suspicious user based on the distance between Alice and Eve, namely d_{AE} , and Eve's computational capability. The value of d_{AE} directly affects the level of SNR at Eve, which further affects the value of

$P_{\text{D},\text{Eve}}$. On the other hand, Eve's computational capability mainly determines the value of $P_{\text{D},\text{Eve}}$. However, the computational capability is difficult to accurately assess in practical situations. For security purpose, a system designer can present a conservative solution by assuming $I_{\text{Eve}} = 1$. Then (21) can be further simplified as

$$\begin{aligned} P_{\text{SA}} &= \max\{P_{\text{D},\text{Bob}} - P_{\text{D},\text{Eve}}, 0\} \\ \text{subject to } &P_{\text{FA},\text{Bob}} \leq \varepsilon_{\text{PFA},\text{Bob}} \\ &P_{\text{FA},\text{Eve}} \leq \varepsilon_{\text{PFA},\text{Eve}} \end{aligned} \quad (22)$$

Apparently, the values of PSA calculated by (22) are more conservative than those calculated by (21).

4 AUTH-SUP SCHEME

In the Auth-SUP scheme [4], the tagged signal can be expressed as

$$\mathbf{x}_i = \rho_s \mathbf{s}_i + \rho_t \mathbf{t}_i, \quad (23)$$

where ρ_s^2 and ρ_t^2 can be interpreted as energy allocations of the message and tag, respectively. According to the result from [4], we obtain $\mathbb{E}\{\mathbf{s}_i^H \mathbf{t}_i\} = 0$, where $\mathbb{E}\{\cdot\}$ represents the operation of extracting the expectation. We have constraint $\rho_s^2 + \rho_t^2 = 1$. Apparently, ρ_s or ρ_t can be regarded as a parameter of the Auth-SUP scheme. In the case where $\rho_t^2 = 0$, the transmitted signal does not contain any authentication tag and $\mathbf{x}_i = \mathbf{s}_i$, which is defined as a normal signal.

4.1 Detection by Eve

Test statistic $\tau_{i,\text{Eve}}$ for the different hypotheses in (16) can be respectively expressed by

$$\tau_{i,\text{Eve}}|H_0 = \mathbf{s}_i^H \mathbf{s}_i - \mathbf{s}_i^H \mathbf{s}_i - v_{i,\text{Eve}} = -v_{i,\text{Eve}}, \quad (24)$$

and

$$\tau_{i,\text{Eve}}|H_1 = (1 - \rho_s) \mathbf{s}_i^H \mathbf{s}_i - \rho_t \mathbf{t}_i^H \mathbf{s}_i - v_{i,\text{Eve}}, \quad (25)$$

where $v_{i,\text{Eve}} = (h_{i,\text{Eve}}^* / |h_{i,\text{Eve}}|^2) \mathbf{s}_i^H \mathbf{n}_{i,\text{Eve}}$ is a zero-mean Gaussian RV with variance $\sigma_{v_{i,\text{Eve}}}^2 = L \sigma_{n,\text{Eve}}^2 / |h_{i,\text{Eve}}|^2 = L / \gamma_{i,\text{Eve}}$, and $\gamma_{i,\text{Eve}} = |h_{i,\text{Eve}}|^2 / \sigma_{n,\text{Eve}}^2$. Because $\mathbb{E}\{\tau_{i,\text{Eve}}|H_0\} = 0$ and $\text{var}\{\tau_{i,\text{Eve}}|H_0\} = L / \gamma_{i,\text{Eve}}$, $\tau_{i,\text{Eve}}|H_0 \sim \mathcal{CN}(0, L / \gamma_{i,\text{Eve}})$. Simultaneously, because $\mathbb{E}\{\tau_{i,\text{Eve}}|H_1\} = (1 - \rho_s)L$ and $\text{var}\{\tau_{i,\text{Eve}}|H_1\} = L / \gamma_{i,\text{Eve}}$, $\tau_{i,\text{Eve}}|H_1 \sim \mathcal{CN}((1 - \rho_s)L, L / \gamma_{i,\text{Eve}})$. Then, because $\delta_{i,\text{Eve}} = \mathbb{R}\{\tau_{i,\text{Eve}}\}$, the threshold test using the hypotheses defined in (15) can be transformed into

$$\begin{aligned} H_0 : & \quad \delta_{i,\text{Eve}} \sim \mathcal{N}\left(0, \frac{L}{2\gamma_{i,\text{Eve}}}\right) \\ H_1 : & \quad \delta_{i,\text{Eve}} \sim \mathcal{N}\left((1 - \rho_s)L, \frac{L}{2\gamma_{i,\text{Eve}}}\right). \end{aligned} \quad (26)$$

Here, the PFA of Eve's detection in the Auth-SUP scheme for the i th block can be expressed as

$$\begin{aligned} P_{\text{FA},i,\text{Eve}} &= \Pr\{\delta_{i,\text{Eve}} > \theta_{i,\text{Eve}}|H_0\} \\ &= Q\left(\frac{\theta_{i,\text{Eve}}}{\sqrt{\text{var}\{\delta_{i,\text{Eve}}|H_0\}}}\right) = Q\left(\theta_{i,\text{Eve}} \sqrt{\frac{2\gamma_{i,\text{Eve}}}{L}}\right). \end{aligned} \quad (27)$$

We utilize the results of ([27], Eq. (5.6)), which is

$$\frac{1}{b} \int_0^\infty Q(\sqrt{x}) \exp\left(\frac{-x}{b}\right) dx = \frac{1}{2} \left(1 - \sqrt{\frac{b}{2+b}}\right), \quad (28)$$

where b is any nonzero constant. Because $h_{i,\text{Eve}}$ is a zero-mean complex Gaussian RV with variance $\sigma_{h,\text{Eve}}^2$, $|h_i|^2$ is an exponential RV. The PFA of a randomly chosen block with a zero-mean complex Gaussian channel realization with optimal threshold θ_{Eve}^0 is thus given by

$$\begin{aligned} P_{\text{FA},\text{Eve}} &= \mathbb{E} \left\{ Q \left(\sqrt{\frac{2(\theta_{\text{Eve}}^0)^2 |h_{i,\text{Eve}}|^2}{L\sigma_{n,\text{Eve}}^2}} \right) \right\} \\ &= \frac{1}{2} \left(1 - \sqrt{\frac{(\theta_{\text{Eve}}^0)^2 \gamma_{\text{Eve}}}{L + (\theta_{\text{Eve}}^0)^2 \gamma_{\text{Eve}}}} \right), \end{aligned} \quad (29)$$

where $\gamma_{\text{Eve}} = \mathbb{E}\{\gamma_{i,\text{Eve}}\} = \sigma_{h,\text{Eve}}^2 / \sigma_{n,\text{Eve}}^2$.

Optimal threshold θ_{Eve}^0 of this test, which is determined for a PFA $\varepsilon_{\text{PFA},\text{Eve}}$, can be obtained as

$$\theta_{\text{Eve}}^0 = \sqrt{\frac{(1 - 2\varepsilon_{\text{PFA},\text{Eve}})^2 L}{4(\varepsilon_{\text{PFA},\text{Eve}} - \varepsilon_{\text{PFA},\text{Eve}}^2) \gamma_{\text{Eve}}}}. \quad (30)$$

Then, the PD of Eve's detection in the Auth-SUP scheme for the i th block can be expressed as

$$\begin{aligned} P_{\text{D},i,\text{Eve}} &= \Pr\{\delta_{i,\text{Eve}} > \theta_{\text{Eve}}^0 | H_1\} \\ &= \begin{cases} Q\left(\sqrt{\frac{2(\theta_{\text{Eve}}^0 - (1-\rho_s)L)^2 \gamma_{i,\text{Eve}}}{L}}\right), & \theta_{\text{Eve}}^0 \geq (1-\rho_s)L \\ 1 - Q\left(\sqrt{\frac{2(\theta_{\text{Eve}}^0 - (1-\rho_s)L)^2 \gamma_{i,\text{Eve}}}{L}}\right), & \theta_{\text{Eve}}^0 < (1-\rho_s)L. \end{cases} \end{aligned} \quad (31)$$

The PD of Eve's detection over a randomly chosen block with a zero-mean complex Gaussian channel realization with optimal threshold θ_{Eve}^0 is thus given by

$$\begin{aligned} P_{\text{D},\text{Eve}} &= \mathbb{E}\{\Pr\{\delta_{i,\text{Eve}} > \theta_{\text{Eve}}^0 | H_1\}\} \\ &= \begin{cases} \frac{1}{2} \left(1 - \sqrt{\frac{(\theta_{\text{Eve}}^0 - (1-\rho_s)L)^2 \gamma_{\text{Eve}}}{L + (\theta_{\text{Eve}}^0 - (1-\rho_s)L)^2 \gamma_{\text{Eve}}}} \right), & \theta_{\text{Eve}}^0 \geq (1-\rho_s)L \\ \frac{1}{2} \left(1 + \sqrt{\frac{(\theta_{\text{Eve}}^0 - (1-\rho_s)L)^2 \gamma_{\text{Eve}}}{L + (\theta_{\text{Eve}}^0 - (1-\rho_s)L)^2 \gamma_{\text{Eve}}}} \right), & \theta_{\text{Eve}}^0 < (1-\rho_s)L. \end{cases} \end{aligned} \quad (32)$$

In summary

$$\begin{aligned} P_{\text{D},\text{Eve}} &= \frac{1}{2} \left(1 - \text{sign}(\theta_{\text{Eve}}^0 - (1-\rho_s)L) \cdot \sqrt{\frac{(\theta_{\text{Eve}}^0 - (1-\rho_s)L)^2 \gamma_{\text{Eve}}}{L + (\theta_{\text{Eve}}^0 - (1-\rho_s)L)^2 \gamma_{\text{Eve}}}} \right), \end{aligned} \quad (33)$$

where $\text{sign}(x)$ in (33) represents a symbol decision function, i.e., $\text{sign}(x) = 1$, if $x \geq 0$; otherwise $\text{sign}(x) = -1$.

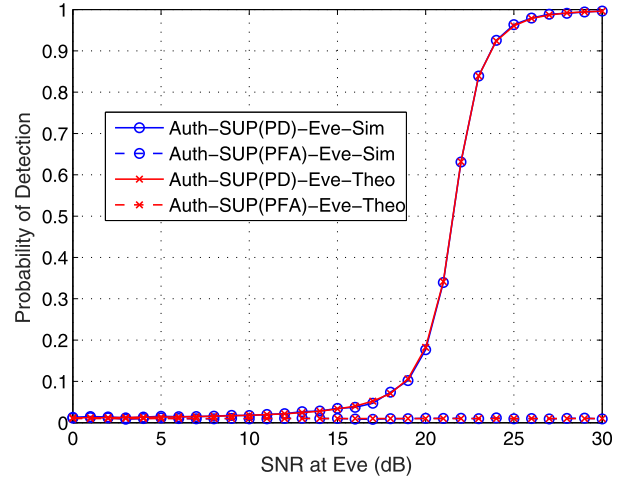


Fig. 2. PD and PFA of the detection at Eve for the Auth-SUP scheme versus different SNRs when the theoretical PFA bound is $\varepsilon_{\text{PFA},\text{Eve}} = 0.01$, where $\rho_t^2 = 0.1$, $L = 64$ and BPSK modulation is adopted.

Fig. 2 shows the plots of the PD and PFA of Eve's detection in the Auth-SUP scheme at various SNRs under $\varepsilon_{\text{PFA},\text{Eve}} = 0.01$, where $\rho_t^2 = 0.1$. In the remaining experiments, we assume that $L = 64$ and binary phase shift keying (BPSK) is adopted, unless otherwise stated. As expected, the closed-form expressions for PD (33) and PFA (29) perfectly match with their Monte-Carlo simulation results. For the Monte-Carlo experiments in this paper, 10^6 independent trials are used to obtain the average results unless otherwise specified.

4.2 Authentication by Bob

Bob's receiver looks for the tag in the residual signal, which is constructed as

$$\mathbf{r}_i = \frac{1}{\rho_t} (\hat{\mathbf{x}}_i - \rho_s \hat{\mathbf{s}}_i). \quad (34)$$

We perform a threshold test using the hypotheses defined in (6), and obtain test statistic $\delta_{i,\text{Bob}}$ by match-filtering the residual with the estimated tag and taking its real part, which is expressed as

$$\delta_{i,\text{Bob}} = \mathbb{R}\{\tau_{i,\text{Bob}}\} = \mathbb{R}\{\mathbf{t}_i^H \mathbf{r}_i\}. \quad (35)$$

According to the results in [4], test statistic $\tau_{i,\text{Bob}}$ for the different hypotheses can be expressed by the following:

For H_0

$$\tau_{i,\text{Bob}}|H_0 = \frac{1}{\rho_t} \left((1-\rho_s) \sum_{k=1}^L t_{i,k}^* s_{i,k} + \frac{h_i^*}{|h_i|^2} \sum_{k=1}^L t_{i,k}^* n_{i,k} \right). \quad (36)$$

Because $\mathbb{E}\{\mathbf{t}_i^H \mathbf{s}_i\} = 0$, we obtain $\mathbb{E}\{\tau_{i,\text{Bob}}|H_0\} = 0$ and $\text{var}\{\tau_{i,\text{Bob}}|H_0\} = L/\rho_t^2 \gamma_{i,\text{Bob}}$, where $\gamma_{i,\text{Bob}} = |h_{i,\text{Bob}}|^2 / \sigma_{n,\text{Bob}}^2$. Thus, $\tau_{i,\text{Bob}}|H_0 \sim \mathcal{CN}(0, L/\rho_t^2 \gamma_{i,\text{Bob}})$.

For H_1

$$\tau_{i,\text{Bob}}|H_1 = \frac{1}{\rho_t} \left(\rho_t \sum_{k=1}^L t_{i,k}^* t_{i,k} + \frac{h_i^*}{|h_i|^2} \sum_{k=1}^L t_{i,k}^* n_{i,k} \right). \quad (37)$$

Subsequently $\mathbb{E}\{\tau_{i,\text{Bob}}|H_1\} = L$ and $\text{var}\{\tau_{i,\text{Bob}}|H_1\} = L/\rho_t^2\gamma_{i,\text{Bob}}$. Thus, $\tau_{i,\text{Bob}}|H_1 \sim \mathcal{CN}(L, L/\rho_t^2\gamma_{i,\text{Bob}})$. Then, the threshold test using the hypotheses defined in (6) can be transformed into

$$\begin{aligned} H_0 : \quad \delta_{i,\text{Bob}} &\sim \mathcal{N}\left(0, \frac{L}{2\rho_t^2\gamma_{i,\text{Bob}}}\right) \\ H_1 : \quad \delta_{i,\text{Bob}} &\sim \mathcal{N}\left(L, \frac{L}{2\rho_t^2\gamma_{i,\text{Bob}}}\right). \end{aligned} \quad (38)$$

Here the PFA of Bob's detection in the Auth-SUP scheme for the i th block can be expressed as

$$\begin{aligned} P_{\text{FA},i,\text{Bob}} &= \Pr\{\delta_{i,\text{Bob}} > \theta_{i,\text{Bob}}|H_0\} \\ &= Q\left(\frac{\theta_{i,\text{Bob}}}{\sqrt{\text{var}\{\delta_{i,\text{Bob}}|H_0\}}}\right) \\ &= Q\left(\sqrt{\frac{2\theta_{i,\text{Bob}}^2\rho_t^2\gamma_{i,\text{Bob}}}{L}}\right). \end{aligned} \quad (39)$$

The PFA of a randomly chosen block with zero-mean complex Gaussian channel realization with optimal threshold θ_{Bob}^0 is thus given by

$$\begin{aligned} P_{\text{FA},\text{Bob}} &= \mathbb{E}\left\{Q\left(\sqrt{\frac{(\theta_{\text{Bob}}^0)^2 2\rho_t^2\gamma_{i,\text{Bob}}}{L}}\right)\right\} \\ &= \frac{1}{2}\left(1 - \sqrt{\frac{(\theta_{\text{Bob}}^0)^2 \rho_t^2\gamma_{\text{Bob}}}{L + (\theta_{\text{Bob}}^0)^2 \rho_t^2\gamma_{\text{Bob}}}}\right), \end{aligned} \quad (40)$$

where $\gamma_{\text{Bob}} = \mathbb{E}\{\gamma_{i,\text{Bob}}\} = \sigma_{h,\text{Bob}}^2/\sigma_{n,\text{Bob}}^2$.

Optimal threshold θ_{Bob}^0 in this test, which is determined for a PFA $\varepsilon_{\text{PFA},\text{Bob}}$, can be obtained as

$$\theta_{\text{Bob}}^0 = \sqrt{\frac{(1 - 2\varepsilon_{\text{PFA},\text{Bob}})^2 L}{4(\varepsilon_{\text{PFA},\text{Bob}} - \varepsilon_{\text{PFA},\text{Bob}}^2)\rho_t^2\gamma_{\text{Bob}}}}. \quad (41)$$

Then, the PD of the Auth-SUP scheme for the i th block can be expressed as

$$\begin{aligned} P_{\text{D},i,\text{Bob}} &= \Pr\{\delta_{i,\text{Bob}} > \theta_{i,\text{Bob}}|H_1\} \\ &= \begin{cases} Q\left(\sqrt{\frac{2(\theta_{i,\text{Bob}} - L)^2 \rho_t^2\gamma_{i,\text{Bob}}}{L}}\right) & \theta_{i,\text{Bob}} \geq L \\ 1 - Q\left(\sqrt{\frac{2(\theta_{i,\text{Bob}} - L)^2 \rho_t^2\gamma_{i,\text{Bob}}}{L}}\right) & \theta_{i,\text{Bob}} < L. \end{cases} \end{aligned} \quad (42)$$

Thus, the PD of a randomly chosen block with a zero-mean complex Gaussian channel realization with optimal threshold θ_{Bob}^0 is given by

$$\begin{aligned} P_{\text{D},\text{Bob}} &= \mathbb{E}\{\Pr\{\delta_{i,\text{Bob}} > \theta_{i,\text{Bob}}|H_1\}\} \\ &= \begin{cases} \frac{1}{2}\left(1 - \sqrt{\frac{(\theta_{\text{Bob}}^0 - L)^2 \rho_t^2\gamma_{\text{Bob}}}{L + (\theta_{\text{Bob}}^0 - L)^2 \rho_t^2\gamma_{\text{Bob}}}}\right), & \theta_{\text{Bob}}^0 \geq L \\ \frac{1}{2}\left(1 + \sqrt{\frac{(\theta_{\text{Bob}}^0 - L)^2 \rho_t^2\gamma_{\text{Bob}}}{L + (\theta_{\text{Bob}}^0 - L)^2 \rho_t^2\gamma_{\text{Bob}}}}\right), & \theta_{\text{Bob}}^0 < L. \end{cases} \end{aligned} \quad (43)$$

In summary

$$\begin{aligned} P_{\text{D},\text{Bob}} &= \frac{1}{2}\left(1 - \text{sign}(\theta_{\text{Bob}}^0 - L)\sqrt{\frac{(\theta_{\text{Bob}}^0 - L)^2 \rho_t^2\gamma_{\text{Bob}}}{L + (\theta_{\text{Bob}}^0 - L)^2 \rho_t^2\gamma_{\text{Bob}}}}\right). \end{aligned} \quad (44)$$

Here we can explicitly obtain the PD of the Auth-SUP scheme in a closed-form. For more details of the conventional Auth-SUP scheme, the readers are please referred to [4].

Fig. 3 shows the plot of the PD and PFA of Bob's authentication for the Auth-SUP scheme for various SNRs under $\varepsilon_{\text{PFA},\text{Bob}} = 0.01$, where $\rho_t^2 = 0.1$. As expected, the closed-form expressions for PD (44) and PFA (40) perfectly match with their Monte-Carlo simulation results. By comparing the PD results shown in Fig. 3 with those in Fig. 2, we can see that the Auth-SUP scheme has a much better PD at Bob than that at Eve, which indicates that an obvious non-zero PSA exists in this example. The PSA analysis for the Auth-SUP scheme is discussed in the next section.

4.3 Performance Analysis of PSA of the Auth-SUP Scheme

Substituting (44) and (33) into (21), then (21) can be rewritten as Eq. (45), shown at the top of the next page.

$$\begin{aligned} P_{\text{SA}} &= \max\{\eta_{\text{Bob}} - \eta_{\text{Eve}}, 0\} \\ &= \max\left\{\frac{1}{2}\left(1 - \text{sign}(\theta_{\text{Bob}}^0 - L)\sqrt{\frac{(\theta_{\text{Bob}}^0 - L)^2 \rho_t^2\gamma_{\text{Bob}}}{L + (\theta_{\text{Bob}}^0 - L)^2 \rho_t^2\gamma_{\text{Bob}}}}\right) \right. \\ &\quad \left. - I_{\text{Eve}}\left(1 - \text{sign}(\theta_{\text{Eve}}^0 - (1 - \rho_s)L)\sqrt{\frac{(\theta_{\text{Eve}}^0 - (1 - \rho_s)L)^2 \gamma_{\text{Eve}}}{L + (\theta_{\text{Eve}}^0 - (1 - \rho_s)L)^2 \gamma_{\text{Eve}}}}\right), 0\right\}. \end{aligned} \quad (45)$$

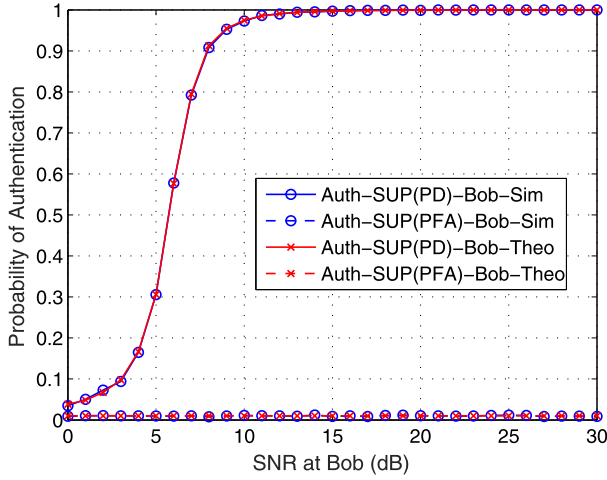


Fig. 3. PD and PFA of the authentication at Bob for the Auth-SUP scheme versus different SNRs when the theoretical PFA bound is $\varepsilon_{\text{PFA,Bob}} = 0.01$, where $\rho_t^2 = 0.1$.

As we know, the performance of a threshold test using two Gaussian hypotheses can be measured by the deflection coefficient [28]. The deflection coefficients under the Auth-SUP scheme for the performance of Bob and Eve's detections are respectively expressed as

$$d_{\text{Bob,SUP}}^2 = \frac{(\mathbb{E}\{\delta_{i,\text{Bob}}|H_1\} - \mathbb{E}\{\delta_{i,\text{Bob}}|H_0\})^2}{\text{var}\{\delta_{i,\text{Bob}}|H_0\}} \quad (46)$$

$$= 2(1 - \rho_s^2)L\gamma_{i,\text{Bob}},$$

and

$$d_{\text{Eve,SUP}}^2 = \frac{(\mathbb{E}\{\delta_{i,\text{Eve}}|H_1\} - \mathbb{E}\{\delta_{i,\text{Eve}}|H_0\})^2}{\text{var}\{\delta_{i,\text{Eve}}|H_0\}} \quad (47)$$

$$= 2(1 - \rho_s)^2L\gamma_{i,\text{Eve}}.$$

From (46) and (47), we can see that $d_{\text{Bob,SUP}}^2$ is proportional to $\gamma_{i,\text{Bob}}$, whereas $d_{\text{Eve,SUP}}^2$ is proportional to $\gamma_{i,\text{Eve}}$. However, L and ρ_s should be carefully set, because both simultaneously affect $d_{\text{Bob,SUP}}^2$ and $d_{\text{Eve,SUP}}^2$. To apply the Auth-SUP scheme into practice, we suggest a range of ρ_s to provide a non-zero PSA, as provided in the following proposition.

Proposition 1. For the Auth-SUP scheme, the block length is L ; Eve's and Bob's SNRs are γ_{Eve} and γ_{Bob} , respectively; and Eve's and Bob's PFA are bounded by $\varepsilon_{\text{PFA,Eve}}$ and $\varepsilon_{\text{PFA,Bob}}$, respectively. The parameter range of the Auth-SUP scheme is given by

$$1 - \sqrt{\frac{(1 - 2\varepsilon_{\text{PFA,Eve}})^2}{4(\varepsilon_{\text{PFA,Eve}} - \varepsilon_{\text{PFA,Eve}}^2)L\gamma_{\text{Eve}}}} \leq \rho_s < \sqrt{1 - \frac{(1 - 2\varepsilon_{\text{PFA,Bob}})^2}{4(\varepsilon_{\text{PFA,Bob}} - \varepsilon_{\text{PFA,Bob}}^2)L\gamma_{\text{Bob}}}}, \quad (48)$$

which is a sufficient but a non-necessary condition to provide a non-zero PSA.

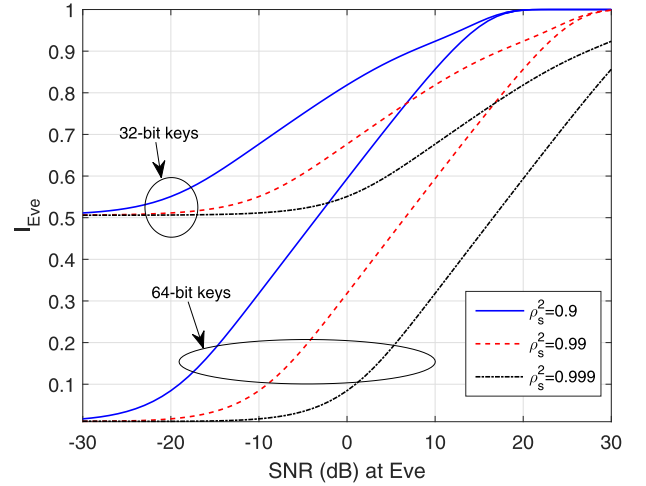


Fig. 4. Ratio of authentication information leakage I_{Eve} versus Eve's SNR under three different ρ_s^2 .

Proof. We consider a situation with $\text{sign}(\theta_{\text{Eve}}^0 - (1 - \rho_s)L) = 1$ and $\text{sign}(\theta_{\text{Bob}}^0 - L) = -1$, where P_{SA} is always non-zero, which is expressed as

$$P_{\text{SA}} = \frac{1}{2} \left((1 - I_{\text{Eve}}) + \sqrt{\frac{(\theta_{\text{Bob}}^0 - L)^2 \rho_t^2 \gamma_{\text{Bob}}}{L + (\theta_{\text{Bob}}^0 - L)^2 \rho_t^2 \gamma_{\text{Bob}}}} \right) \quad (49)$$

$$I_{\text{Eve}} \sqrt{\frac{(\theta_{\text{Eve}}^0 - (1 - \rho_s)L)^2 \gamma_{\text{Eve}}}{L + (\theta_{\text{Eve}}^0 - (1 - \rho_s)L)^2 \gamma_{\text{Eve}}}},$$

because

$$\theta_{\text{Eve}}^0 = \sqrt{\frac{(1 - 2\varepsilon_{\text{PFA,Eve}})^2 L}{4(\varepsilon_{\text{PFA,Eve}} - \varepsilon_{\text{PFA,Eve}}^2)\gamma_{\text{Eve}}}} \geq (1 - \rho_s)L \quad (50)$$

$$\rightarrow \rho_s \geq 1 - \sqrt{\frac{(1 - 2\varepsilon_{\text{PFA,Eve}})^2}{4(\varepsilon_{\text{PFA,Eve}} - \varepsilon_{\text{PFA,Eve}}^2)L\gamma_{\text{Eve}}}},$$

and

$$\theta_{\text{Bob}}^0 = \sqrt{\frac{(1 - 2\varepsilon_{\text{PFA,Bob}})^2 L}{4(\varepsilon_{\text{PFA,Bob}} - \varepsilon_{\text{PFA,Bob}}^2)\rho_t^2 \gamma_{\text{Bob}}}} < L \quad (51)$$

$$\rightarrow \rho_s < \sqrt{1 - \frac{(1 - 2\varepsilon_{\text{PFA,Bob}})^2}{4(\varepsilon_{\text{PFA,Bob}} - \varepsilon_{\text{PFA,Bob}}^2)L\gamma_{\text{Bob}}}}.$$

Since $0 \leq I_{\text{Eve}} \leq 1$, we obtain $(1 - I_{\text{Eve}}) \geq 0$. Then the results of (49) are always larger than zero, and the proof of this proposition is then completed. $\square$

An example of I_{Eve} as a function of Eve's SNR is shown in Fig. 4. The curves are parameterized by the key length (32-bit versus 64-bit) while the tag length remains fixed at 64 bits. We use (24) and (25) to calculate the conditional key distribution. First, if the distance between Alice and Eve is very far, Eve's SNR is very low because $\sigma_h^2 \propto d^{-\alpha_d}$. We note that when Eve has a noisier observation due to low Eve's SNR, higher key equivocation would result, and the value

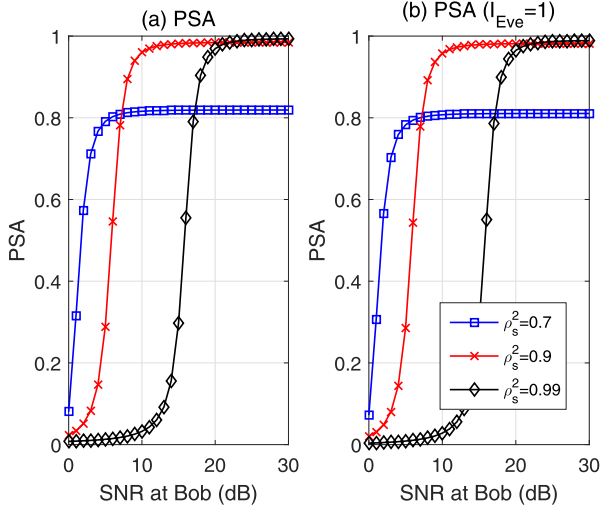


Fig. 5. PSA of the Auth-SUP scheme versus different Bob's SNRs under three different ρ_s^2 when the theoretical PFA bound is $\varepsilon_{\text{PFA,Bob}} = \varepsilon_{\text{PFA,Eve}} = 0.01$, where $\gamma_{\text{Eve}} = 10$ dB, (a) I_{Eve} is calculated by (20); (b) $I_{\text{Eve}} = 1$.

of I_{Eve} is reduced. Second, decreasing the key length relative to the tag length decreases the equivocation and increases the value of I_{Eve} . This is the same conclusion as that in [21], because when the valid tags become sparser in the tag space $\mathcal{T}$, conditional distribution (13) becomes less uniform; thus, the key equivocation decreases, and the value of I_{Eve} increases. The value of I_{Eve} reaches zero when Eve's SNR is sufficiently low. In this case, Eve cannot make any threat to the system because she is too far away from Alice.

Fig. 5 shows the plots of the PSA of the Auth-SUP scheme for various Bob's SNRs under three different ρ_s^2 values, when $\varepsilon_{\text{PFA,Bob}} = \varepsilon_{\text{PFA,Eve}} = 0.01$, where $\gamma_{\text{Eve}} = 10$ dB. In Fig. 5a, I_{Eve} is calculated by (20), and in Fig. 5b $I_{\text{Eve}} = 1$. Both subfigures in Fig. 5 show that $\rho_s^2 = 0.9$ is the best option among the three cases, whereas the PSA performance deteriorates when ρ_s^2 either increases or decreases. Thus, a tradeoff for setting ρ_s^2 is possible.

We note that, in the considered system model, Bob's SNR and Eve's SNR are determined by their corresponding distances to Alice, respectively. If Eve is closer to Alice than Bob, Eve's SNR is of course better than Bob's SNR. We indeed cannot control Eve's location. However, if Alice is aware of a suspicious user coming closer, she assesses the security level through the proposed security model. If she finds the security level below some threshold, a wise option for Alice is to adjust its authentication parameters, or even to cancel this authentication request to await an appropriate chance.

By comparing Figs. 5a and 5b, their results are very similar, and the effect of I_{Eve} appears that it can be neglected. However, this is not true. The values of PSA shown in Fig. 5b are inferior to those in Fig. 5a, as we expected in Section 3. Since both $P_{\text{D,Eve}}$ and I_{Eve} are independent of Bob's SNRs, γ_{Bob} , they have constant values in Fig. 5. Specifically, for $\rho_s^2 = 0.7, 0.9$, and 0.99 , $P_{\text{D,Eve}} = 0.1895, 0.0180$ and 0.0105 , respectively, whereas $I_{\text{Eve}} = 0.9551, 0.8565$ and 0.5936 , respectively. Since the value of $\eta_{\text{Eve}} = P_{\text{D,Eve}} I_{\text{Eve}}$ becomes very small in this example, the value of P_{SA} is determined by $P_{\text{D,Bob}}$, and the effect of I_{Eve} on P_{SA} becomes more obvious as the value of $P_{\text{D,Bob}}$ decreases. We consider $\rho_s^2 = 0.99$ as an example to

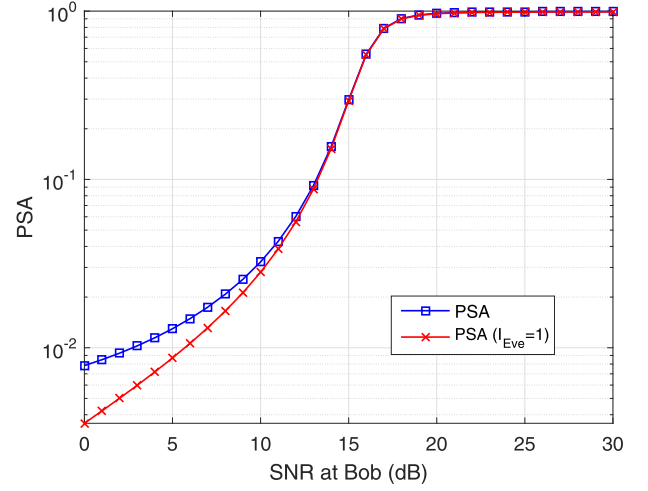


Fig. 6. PSA of the Auth-SUP scheme versus different Bob's SNRs under different I_{Eve} , where the remaining parameters are the same as those in Fig. 5 except $\rho_s^2 = 0.99$.

illustrate the effect of I_{Eve} on P_{SA} , as shown in Fig. 6. Fig. 6 clearly shows that the values of the PSA calculated by (22) are inferior to those calculated by (21), but the difference becomes obvious for $\gamma_{\text{Bob}} \leq 10$ dB. Note that $\gamma_{\text{Eve}} = 10$ dB in this example. This implies that when a suspicious user is closer to Alice than Bob to Alice, i.e., $d_{\text{AE}} \leq d_{\text{AB}}$, which also indicates that $\gamma_{\text{Bob}} \leq \gamma_{\text{Eve}}$, the system designer can assess Eve's computational capability for estimating the value of I_{Eve} if possible, and re-design the parameters of certain PHY-layer authentication ϕ_i to further improve the value of P_{SA} .

To improve the practical value of the proposed security model, in the following sections of this paper, we present our calculation of the value of P_{SA} using (22) rather than (21). The reasons of this decision are twofold. First, assessing Eve's computational capability in practical situations is very difficult, since Eve acting as an adversary, can easily hide her actual capability. Second, more importantly, the value of P_{SA} calculated using (22) is a conservative solution by taking security as the first concern. Now, the proposed security model becomes a tradeoff process between two detection mechanisms, where one has side-information and one does not have it, which is an easy implementation.

5 AUTH-REP SCHEME

The other PHY-layer authentication schemes [5], [19], [29] obviously violate the requirement of the proposed security model. The reason behind this is that these schemes break the original property of the normal signal, which make the PD at Eve $P_{\text{D,Eve}}$ high and make its corresponding PSA P_{SA} very low. For example, the Auth-TDM scheme modifies the original frame structure by filling it with extra transmission symbols, as shown in Fig. 7a, and the scheme in [5] increases the number of constellation points compared with that in the normal signal, which allows an adversary to easily detect the authentication tag through these anomalous features, and consequently sacrifices their security.

In this section, we proposed a variant of the Auth-TDM scheme, as shown in Fig. 7a for performance comparison. This proposed scheme directly replaces some symbols of the normal signal by those of the tag signal rather than filling them with extra symbols. This variant is referred to as

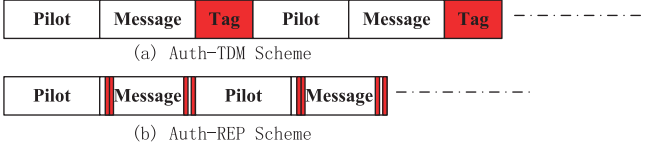


Fig. 7. Frame structure of the Auth-TDM and Auth-REP schemes.

the Auth-REP scheme and is shown in Fig. 7b. We note that the method of embedding tag in the Auth-REP scheme is the same as that in [2], but they have different ways of generating the authentication tag. In [2], the authentication tag was generated using one-way hash chains rather than based on a secret key. The authentication scheme in [2] was applied into a cognitive radio network to fight primary user emulation (PUE) attacks, which can be regarded as broadcasting authentication.

In the Auth-REP scheme, secret key $\mathbf{k}$ not only generates tag signal $\mathbf{t}_i$ but also controls the position of embedded $\mathbf{t}_i$ into the message cluster of the normal signal. More specifically, we consider the output of channel coding $\mathbf{c}_i$ as a target carrier. Alice randomly selects L_R bits of $\mathbf{c}_i$ to be embedded to $\mathbf{t}_i$. Apparently, the length of $\mathbf{t}_i$ in the Auth-SUP scheme is L , whereas that in the Auth-REP scheme is L_R . Whenever we embed a tag bit, we only replace the bit if it differs from the tag bit. Because both $\mathbf{c}_i$ and $\mathbf{t}_i$ are random sequence, the possibility of a certain bit to be modified is $1/2$. Then, without loss of generality, we assume that $L_R/2$ bits of $\mathbf{c}_i$ are available for modification in the Auth-REP scheme, which is denoted as $\mathbf{c}_{i,\text{REP}}$. Through modulation, $\mathbf{c}_{i,\text{REP}}$ is modulated as $\mathbf{s}_{i,\text{REP}}$. Then, the tagged signal of the Auth-REP scheme can be represented as $\mathbf{x}_i = \mathbf{s}_{i,\text{REP}}$. For $L_R = 0$, the transmitted signal does not contain any authentication tag, and $\mathbf{x}_i = \mathbf{s}_i$, which is a normal signal. The portion of the tag signal in the transmitted signal is denoted as $\mathbf{x}_i$, i.e., for a tagged signal $\mathbf{x}_i = \mathbf{s}_{i,\text{REP}}$, as shown by the red color part in Fig. 7b. For a normal signal, $\mathbf{x}_i = \mathbf{s}_i$.

We note that L_R is a parameter of the Auth-REP scheme and its value should be very small, i.e., $L_R \ll L$, because it obviously avoids destruction of the decoding result of the normal signal, similar to the requirement of ρ_t in the Auth-SUP scheme. In other words, the errors caused by the Auth-REP or Auth-SUP scheme plus those caused by the channel are not beyond the capability of error correction providing by modulation and channel coding.

5.1 Detection by Eve

Test statistics $\tau_{i,\text{Eve}}$ for the different hypotheses in (16) can be respectively expressed as

$$\tau_{i,\text{Eve}}|H_0 = \mathbf{s}_i^H \mathbf{s}_i - \mathbf{s}_i^H \mathbf{s}_i - v_{i,\text{Eve}} = -v_{i,\text{Eve}}, \quad (52)$$

and

$$\tau_{i,\text{Eve}}|H_1 = \mathbf{s}_i^H \mathbf{s}_i - \mathbf{s}_i^H \mathbf{s}_{i,\text{REP}} - v_{i,\text{Eve}}, \quad (53)$$

where $v_{i,\text{Eve}} = (h_{i,\text{Eve}}^*/|h_{i,\text{Eve}}|)^2 \mathbf{s}_i^H \mathbf{n}_{i,\text{Eve}}$ is a zero-mean Gaussian RV with variance $\sigma_{v_{i,\text{Eve}}}^2 = L\sigma_{n,\text{Eve}}^2/|h_{i,\text{Eve}}|^2 = L/\gamma_{i,\text{Eve}}$. Because $\mathbb{E}\{\tau_{i,\text{Eve}}|H_0\} = 0$ and $\text{var}\{\tau_{i,\text{Eve}}|H_0\} = L/\gamma_i$, $\tau_{i,\text{Eve}}|H_0 \sim \mathcal{CN}(0, L/\gamma_{i,\text{Eve}})$. Simultaneously, because $\mathbb{E}\{\tau_{i,\text{Eve}}|H_1\} = L_R$ and $\text{var}\{\tau_{i,\text{Eve}}|H_1\} = L/\gamma_{i,\text{Eve}}$, $\tau_{i,\text{Eve}}|H_1 \sim$

$\mathcal{CN}(L_R, L/\gamma_{i,\text{Eve}})$. Then, the threshold test using the hypotheses defined in (15) can be transformed into

$$\begin{aligned} H_0 : \quad \delta_{i,\text{Eve}} &\sim \mathcal{N}\left(0, \frac{L}{2\gamma_{i,\text{Eve}}}\right) \\ H_1 : \quad \delta_{i,\text{Eve}} &\sim \mathcal{N}\left(L_R, \frac{L}{2\gamma_{i,\text{Eve}}}\right). \end{aligned} \quad (54)$$

Here the PFA of Eve's detection in the Auth-SUP scheme for the i th block can be expressed as

$$P_{\text{FA},i,\text{Eve}} = \Pr\{\delta_{i,\text{Eve}} > \theta_{i,\text{Eve}}|H_0\} = Q\left(\sqrt{\frac{2\theta_{i,\text{Eve}}^2}{L\gamma_{i,\text{Eve}}}}\right). \quad (55)$$

The PFA of a randomly chosen block with a zero-mean complex Gaussian channel realization with optimal threshold θ_{Eve}^0 is given by

$$\begin{aligned} P_{\text{FA},\text{Eve}} &= \mathbb{E}\left\{Q\left(\sqrt{\frac{2(\theta_{\text{Eve}}^0)^2}{L\gamma_{i,\text{Eve}}}}\right)\right\} \\ &= \frac{1}{2}\left(1 - \sqrt{\frac{(\theta_{\text{Eve}}^0)^2 \gamma_{i,\text{Eve}}}{L + (\theta_{\text{Eve}}^0)^2 \gamma_{i,\text{Eve}}}}\right). \end{aligned} \quad (56)$$

Optimal threshold θ_{Eve}^0 of this test, which is determined for PFA $\varepsilon_{\text{PFA},\text{Eve}}$, can be obtained as

$$\theta_{\text{Eve}}^0 = \sqrt{\frac{(1 - 2\varepsilon_{\text{PFA},\text{Eve}})^2 L}{4(\varepsilon_{\text{PFA},\text{Eve}} - \varepsilon_{\text{PFA},\text{Eve}}^2) \gamma_{i,\text{Eve}}}}. \quad (57)$$

Then, the PD of Eve's detection in the Auth-REP scheme for the i th block can be expressed as

$$\begin{aligned} P_{\text{D},i,\text{Eve}} &= \Pr\{\delta_{i,\text{Eve}} > \theta_{\text{Eve}}^0|H_1\} \\ &= \begin{cases} Q\left(\sqrt{\frac{2(\theta_{\text{Eve}}^0 - L_R)^2 \gamma_{i,\text{Eve}}}{L}}\right), & \theta_{\text{Eve}}^0 \geq L_R \\ 1 - Q\left(\sqrt{\frac{2(\theta_{\text{Eve}}^0 - L_R)^2 \gamma_{i,\text{Eve}}}{L}}\right), & \theta_{\text{Eve}}^0 < L_R \end{cases}. \end{aligned} \quad (58)$$

The PD of Eve's detection over a randomly chosen block with a zero-mean complex Gaussian channel realization with optimal threshold θ_{Eve}^0 is thus given by

$$\begin{aligned} P_{\text{D},\text{Eve}} &= \mathbb{E}\{\Pr\{\delta_{i,\text{Eve}} > \theta_{\text{Eve}}^0|H_1\}\} \\ &= \begin{cases} \frac{1}{2}\left(1 - \sqrt{\frac{(\theta_{\text{Eve}}^0 - L_R)^2 \gamma_{\text{Eve}}}{L + (\theta_{\text{Eve}}^0 - L_R)^2 \gamma_{\text{Eve}}}}\right), & \theta_{\text{Eve}}^0 \geq L_R \\ \frac{1}{2}\left(1 + \sqrt{\frac{(\theta_{\text{Eve}}^0 - L_R)^2 \gamma_{\text{Eve}}}{L + (\theta_{\text{Eve}}^0 - L_R)^2 \gamma_{\text{Eve}}}}\right), & \theta_{\text{Eve}}^0 < L_R \end{cases}. \end{aligned} \quad (59)$$

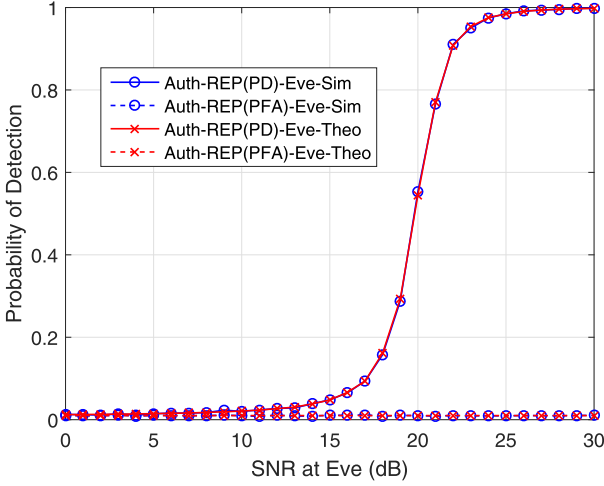


Fig. 8. PD and PFA of the detection at Eve for the Auth-REP scheme versus different SNRs when the theoretical PFA bound is $\varepsilon_{\text{PFA,Eve}} = 0.01$, where $L_R = 4$.

In summary

$$P_{\text{D,Eve}} = \frac{1}{2} \left(1 - \text{sign}(\theta_{\text{Eve}}^0 - L_R) \sqrt{\frac{(\theta_{\text{Eve}}^0 - L_R)^2 \gamma_{\text{Eve}}}{L + (\theta_{\text{Eve}}^0 - L_R)^2 \gamma_{\text{Eve}}}} \right). \quad (60)$$

Fig. 8 shows the plots of the PD and PFA of Eve's detection in the Auth-REP scheme at various SNRs under $\varepsilon_{\text{PFA,Eve}} = 0.01$, where $L_R = 4$. As expected, the closed-form expressions for PD (60) and PFA (56) perfectly match with their Monte-Carlo simulation results.

5.2 Authentication by Bob

Bob's receiver looks for the tag in the residual signal, denoted as $\mathbf{r}_i = \hat{\mathbf{x}}_i$, which is extracted from $\hat{\mathbf{x}}_i$ in (4) along the same position defined by the same secret key $\mathbf{k}$.

We perform a threshold test using the hypotheses defined in (6), and obtain test statistic δ_i by match-filtering the residual with the estimated tag and extracting its real part, which is expressed as

$$\delta_{i,\text{Bob}} = \mathbb{R}\{\tau_{i,\text{Bob}}\} = \mathbb{R}\{\mathbf{s}_{i,\text{REP}}^H \mathbf{r}_i\}. \quad (61)$$

Test statistic $\tau_{i,\text{Bob}}$ for different hypotheses can be expressed by the following.

For H_0 , because $\mathbb{E}\{\mathbf{s}_{i,\text{REP}}^H \mathbf{s}_i\} = 0$, we obtain $\mathbb{E}\{\tau_{i,\text{Bob}}|H_0\} = 0$ and $\text{var}\{\tau_{i,\text{Bob}}|H_0\} = L_R/\gamma_{i,\text{Bob}}$. Thus, $\tau_{i,\text{Bob}}|H_0 \sim \mathcal{CN}(0, L_R/\gamma_{i,\text{Bob}})$.

For H_1 , $\mathbb{E}\{\tau_{i,\text{Bob}}|H_1\} = L_R$, and $\text{var}\{\tau_{i,\text{Bob}}|H_1\} = L_R/\gamma_{i,\text{Bob}}$. Thus, $\tau_{i,\text{Bob}}|H_1 \sim \mathcal{CN}(L_R, L_R/\gamma_{i,\text{Bob}})$.

Then, the threshold test using hypotheses defined in (6) can be transformed into

$$\begin{aligned} H_0 : \quad \delta_{i,\text{Bob}} &\sim \mathcal{N}\left(0, \frac{L_R}{2\gamma_{i,\text{Bob}}}\right) \\ H_1 : \quad \delta_{i,\text{Bob}} &\sim \mathcal{N}\left(L_R, \frac{L_R}{2\gamma_{i,\text{Bob}}}\right). \end{aligned} \quad (62)$$

Here, the PFA of the Auth-REP method for the i th block can be expressed as

$$\begin{aligned} P_{\text{FA},i,\text{Bob}} &= \Pr\{\delta_{i,\text{Bob}} > \theta_{i,\text{Bob}}|H_0\} \\ &= Q\left(\frac{\theta_{i,\text{Bob}}}{\sqrt{\text{var}\{\delta_{i,\text{Bob}}|H_0\}}}\right) = Q\left(\sqrt{\frac{2\theta_{i,\text{Bob}}^2}{L_R\gamma_{i,\text{Bob}}}}\right). \end{aligned} \quad (63)$$

The PFA of a randomly chosen block with zero-mean complex Gaussian channel realization with optimal threshold θ_{Bob}^0 is thus given as

$$\begin{aligned} P_{\text{FA,Bob}} &= \mathbb{E}\left\{Q\left(\sqrt{\frac{2\theta_{i,\text{Bob}}^2 \gamma_{i,\text{Bob}}}{L_R}}\right)\right\} \\ &= \frac{1}{2} \left(1 - \sqrt{\frac{(\theta_{\text{Bob}}^0)^2 \gamma_{\text{Bob}}}{L_R + (\theta_{\text{Bob}}^0)^2 \gamma_{\text{Bob}}}}\right). \end{aligned} \quad (64)$$

Optimal threshold θ_{Bob}^0 of this test, which is determined for PFA $\varepsilon_{\text{PFA,Bob}}$, can be obtained as

$$\theta_{\text{Bob}}^0 = \sqrt{\frac{(1 - 2\varepsilon_{\text{PFA,Bob}})^2 L_R}{4(\varepsilon_{\text{PFA,Bob}} - \varepsilon_{\text{PFA,Bob}}^2) \gamma_{\text{Bob}}}}. \quad (65)$$

Then, the PD of the Auth-SUP scheme for the i th block can be expressed as

$$\begin{aligned} P_{\text{D},i,\text{Bob}} &= \Pr\{\delta_{i,\text{Bob}} > \theta_{i,\text{Bob}}|H_1\} \\ &= \begin{cases} Q\left(\sqrt{\frac{2(\theta_{i,\text{Bob}} - L_R)^2 \gamma_{i,\text{Bob}}}{L_R}}\right) & \theta_{i,\text{Bob}} \geq L_R \\ 1 - Q\left(\sqrt{\frac{2(\theta_{i,\text{Bob}} - L_R)^2 \gamma_{i,\text{Bob}}}{L_R}}\right) & \theta_{i,\text{Bob}} < L_R. \end{cases} \end{aligned} \quad (66)$$

The PD of a randomly chosen block with a zero-mean complex Gaussian channel realization with optimal threshold θ_{Bob}^0 is thus given as

$$\begin{aligned} P_{\text{D,Bob}} &= \mathbb{E}\{\Pr\{\delta_{i,\text{Bob}} > \theta_{i,\text{Bob}}|H_1\}\} \\ &= \begin{cases} \frac{1}{2} \left(1 - \sqrt{\frac{(\theta_{\text{Bob}}^0 - L_R)^2 \sigma_{h,\text{Bob}}^2}{L_R \sigma_{n,\text{Bob}}^2 + (\theta_{\text{Bob}}^0 - L_R)^2 \sigma_{h,\text{Bob}}^2}}\right), & \theta_{\text{Bob}}^0 \geq L_R \\ \frac{1}{2} \left(1 + \sqrt{\frac{(\theta_{\text{Bob}}^0 - L_R)^2 \sigma_{h,\text{Bob}}^2}{L_R \sigma_{n,\text{Bob}}^2 + (\theta_{\text{Bob}}^0 - L_R)^2 \sigma_{h,\text{Bob}}^2}}\right), & \theta_{\text{Bob}}^0 < L_R. \end{cases} \end{aligned} \quad (67)$$

In summary

$$P_{\text{D,Bob}} = \frac{1}{2} \left(1 - \text{sign}(\theta_{\text{Bob}}^0 - L_R) \sqrt{\frac{(\theta_{\text{Bob}}^0 - L_R)^2 \gamma_{\text{Bob}}}{L_R + (\theta_{\text{Bob}}^0 - L_R)^2 \gamma_{\text{Bob}}}}\right). \quad (68)$$

Fig. 9 shows the plots of the PD and PFA of Bob's authentication in the Auth-REP scheme for various SNRs under $\varepsilon_{\text{PFA,Bob}} = 0.01$, where $L_R = 4$. As expected, the closed-form

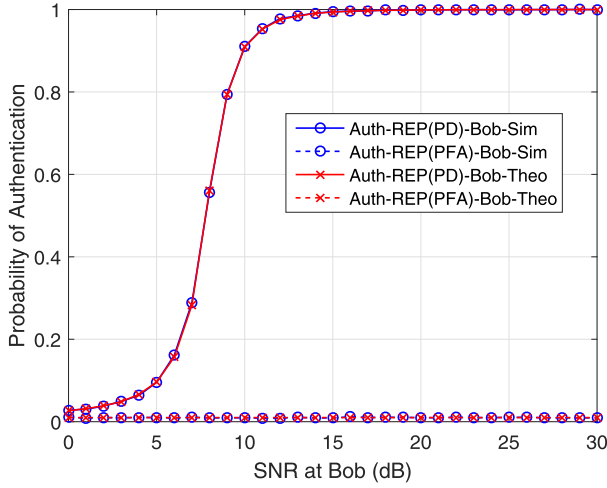


Fig. 9. PD and PFA of the authentication at Bob for the Auth-REP scheme versus different SNRs when the theoretical PFA bound is $\varepsilon_{\text{PFA,Bob}} = 0.01$, where $L_R = 4$.

expressions for PD (68) and PFA (64) perfectly match with their Monte-Carlo simulation results. By comparing the PD results shown in Fig. 9 with those in Fig. 8, we can see that the Auth-REP scheme has a much better PD at Bob than that at Eve, which indicates an obvious presence of non-zero PSA in this example. The PSA analysis for the Auth-REP scheme is discussed in the next section.

5.3 Performance Analysis of PSA of the Auth-REP Scheme

Substituting (68) and (60) into (22) yields (69), shown at the top of the next page

$$P_{\text{D,Secure}} = \max\{P_{\text{D,Bob}} - P_{\text{D,Eve}}, 0\} \\ = \max\left\{\frac{1}{2} \left(\text{sign}(\theta_{\text{Eve}}^0 - L_R) \sqrt{\frac{(\theta_{\text{Eve}}^0 - L_R)^2 \gamma_{\text{Eve}}}{L + (\theta_{\text{Eve}}^0 - L_R)^2 \gamma_{\text{Eve}}}} - \text{sign}(\theta_{\text{Bob}}^0 - L_R) \sqrt{\frac{(\theta_{\text{Bob}}^0 - L_R)^2 \gamma_{\text{Bob}}}{L_R + (\theta_{\text{Bob}}^0 - L_R)^2 \gamma_{\text{Bob}}}} \right), 0 \right\}. \quad (69)$$

The deflection coefficient under the Auth-REP scheme for the performance of Bob's and Eve's detection are respectively expressed as

$$d_{\text{Bob,REP}}^2 = \frac{(\mathbb{E}\{\delta_{i,\text{Bob}}|H_1\} - \mathbb{E}\{\delta_{i,\text{Bob}}|H_0\})^2}{\text{var}\{\delta_{i,\text{Bob}}|H_0\}} \quad (70) \\ = 2L_R \gamma_{i,\text{Bob}},$$

and

$$d_{\text{Eve,REP}}^2 = \frac{(\mathbb{E}\{\delta_{i,\text{Eve}}|H_1\} - \mathbb{E}\{\delta_{i,\text{Eve}}|H_0\})^2}{\text{var}\{\delta_{i,\text{Eve}}|H_0\}} \quad (71) \\ = \frac{2L_R^2 \gamma_{i,\text{Eve}}}{L}.$$

From (70) and (71), we can observe that $d_{\text{Bob,REP}}^2$ is proportional to $\gamma_{i,\text{Bob}}$, whereas $d_{\text{Eve,REP}}^2$ is proportional to $\gamma_{i,\text{Eve}}$. $d_{\text{Bob,REP}}^2$ is independent of L , whereas $d_{\text{Eve,REP}}^2$ is inversely proportional to L . Thus, $P_{\text{D,Secure}}$ of the Auth-REP scheme obviously much more improves as L increases compared with that of the Auth-SUP scheme, which will be analyzed

in the next section. However, L_R should be carefully set, because it simultaneously affects $d_{\text{Bob,SUP}}^2$ and $d_{\text{Eve,SUP}}^2$. To apply the Auth-REP scheme into practice, we suggest a range of ρ_s to provide a non-zero PSA, as presented in the following proposition.

Proposition 2. For the Auth-REP scheme, the block length is L ; Eve's and Bob's SNRs are γ_{Eve} and γ_{Bob} , respectively; and Eve's and Bob's PFA are bounded by $\varepsilon_{\text{PFA,Eve}}$ and $\varepsilon_{\text{PFA,Bob}}$, respectively. The parameter range of the Auth-REP scheme is given as

$$\frac{(1 - 2\varepsilon_{\text{PFA,Bob}})^2}{4(\varepsilon_{\text{PFA,Bob}} - \varepsilon_{\text{PFA,Bob}}^2)\gamma_{\text{Bob}}} < L_R \\ \leq \sqrt{\frac{(1 - 2\varepsilon_{\text{PFA,Eve}})^2 L}{4(\varepsilon_{\text{PFA,Eve}} - \varepsilon_{\text{PFA,Eve}}^2)\gamma_{\text{Eve}}}}, \quad (72)$$

which is a sufficient but a non-necessary condition to provide a non-zero PSA.

Proof. The proof is similar to that of Proposition 1. We consider a situation with $\text{sign}(\theta_{\text{Eve}}^0 - L_R) = 1$ and $\text{sign}(\theta_{\text{Bob}}^0 - L_R) = -1$, where $P_{\text{D,Secure}}$ is always positive, and expressed as

$$P_{\text{D,Secure}} = \frac{1}{2} \left(\sqrt{\frac{(\theta_{\text{Eve}}^0 - L_R)^2 \gamma_{\text{Eve}}}{L + (\theta_{\text{Eve}}^0 - L_R)^2 \gamma_{\text{Eve}}}} + \sqrt{\frac{(\theta_{\text{Bob}}^0 - L_R)^2 \gamma_{\text{Bob}}}{L_R + (\theta_{\text{Bob}}^0 - L_R)^2 \gamma_{\text{Bob}}}} \right). \quad (73)$$

Because

$$\theta_{\text{Eve}}^0 = \sqrt{\frac{(1 - 2\varepsilon_{\text{PFA,Eve}})^2 L}{4(\varepsilon_{\text{PFA,Eve}} - \varepsilon_{\text{PFA,Eve}}^2)\gamma_{\text{Eve}}}} \geq L_R, \quad (74)$$

and

$$\theta_{\text{Bob}}^0 = \sqrt{\frac{(1 - 2\varepsilon_{\text{PFA,Bob}})^2 L_R}{4(\varepsilon_{\text{PFA,Bob}} - \varepsilon_{\text{PFA,Bob}}^2)\gamma_{\text{Bob}}}} < L_R \\ L_R > \frac{(1 - 2\varepsilon_{\text{PFA,Bob}})^2}{4(\varepsilon_{\text{PFA,Bob}} - \varepsilon_{\text{PFA,Bob}}^2)\gamma_{\text{Bob}}}, \quad (75)$$

the proof is then completed. $\square$

Fig. 10 shows that the plots of the PSA of the Auth-SUP scheme for various Bob's SNRs under three different L_R , when $\varepsilon_{\text{PFA,Bob}} = \varepsilon_{\text{PFA,Eve}} = 0.01$, where $\gamma_{\text{Eve}} = 10$ dB. Fig. 10 shows that when the value of L_R increases, the robustness

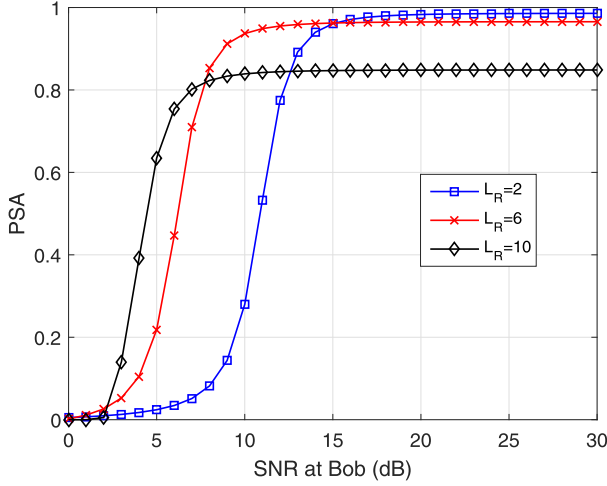


Fig. 10. PSA of the Auth-REP scheme versus different Bob's SNRs under three different L_R when the theoretical PFA bound is $\varepsilon_{\text{PFA,Bob}} = \varepsilon_{\text{PFA,Eve}} = 0.01$, where $\gamma_{\text{Eve}} = 10$ dB.

of Auth-REP scheme is enhanced, especially in Bob's low SNR region, whereas its corresponding security is weakened because it can be more easily detected, especially in Bob's high SNR region. Fig. 10 shows that $L_R = 6$ may be the best option among the three cases, whereas the PSA performance deteriorates when L_R either increases or decreases. Thus, a tradeoff for setting L_R is possible.

6 COMPARISON BETWEEN THE AUTH-SUP AND AUTH-REP SCHEMES

This section presents a comparison of the final performance of the Auth-SUP and Auth-REP schemes in terms of the PSA based on the proposed generic security model. Because the value of $P_{\text{D,Secure}}$ is determined by the difference between $P_{\text{D,Bob}}$ and $P_{\text{D,Eve}}$, we first make $P_{\text{D,Bob}}$ of the two schemes equal by adjusting their own parameters. Then, we compare their PSAs by analyzing $P_{\text{D,Eve}}$ values. The larger $P_{\text{D,Eve}}$ value is, the lesser is P_{SA} , and vice versa. Then we can present the following proposition.

Proposition 3. *For the Auth-SUP and Auth-REP schemes, if we let $P_{\text{D,Bob}}$ of the two schemes equal by adjusting their own parameters, $P_{\text{D,Eve}}$ of the Auth-REP scheme is larger than that of the Auth-SUP scheme. Then P_{SA} of the Auth-SUP scheme is better than that of the Auth-REP scheme.*

Proof. According to (46) and (70), by setting $d_{\text{Bob,SUP}}^2 = d_{\text{Bob,REP}}^2$, we obtain

$$\rho_t^2 = 1 - \rho_s^2 = \frac{L_R}{L}. \quad (76)$$

By substituting (76) into (71), we obtain

$$d_{\text{Eve,REP}}^2 = 2(1 - \rho_s)^2(1 + \rho_s)^2 L \gamma_{i,\text{Eve}}. \quad (77)$$

By comparing (77) with (47), we can draw the conclusion that $d_{\text{Eve,REP}}^2 \geq d_{\text{Eve,SUP}}^2$, because $\rho_s \geq 0$. The equal sign of the inequality is realized only when $\rho_s = 0$, which indicates that the normal signal is totally replaced by the tag and is unreasonable. This inequality indicates that $P_{\text{D,Eve}}$ of the Auth-REP scheme is larger than that of the Auth-

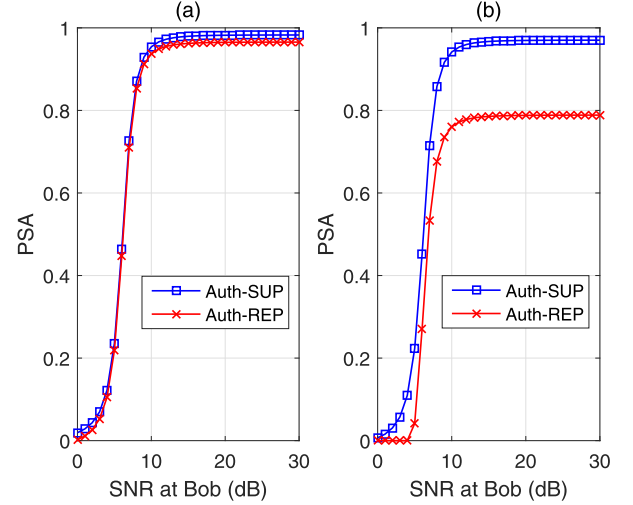


Fig. 11. PSA comparison between the Auth-SUP and Auth-REP schemes versus different Bob's SNRs under different γ_{Eve} when the theoretical PFA bound is $\varepsilon_{\text{PFA,Bob}} = \varepsilon_{\text{PFA,Eve}} = 0.01$, where $L_R = 6$ and $\rho_t^2 = L_R/L$: (a) $\gamma_{\text{Eve}} = 10$ dB and (b) $\gamma_{\text{Eve}} = 15$ dB.

SUP scheme. Then the corresponding P_{SA} of the Auth-SUP scheme is better than that of the Auth-REP scheme. The proof is completed. $\square$

From the above proposition, we can obtain some important observations.

Corollary 1. *Because $\gamma_{i,\text{Eve}}$ only affects $d_{\text{Eve,REP}}^2$ and $d_{\text{Eve,SUP}}^2$, rather than $d_{\text{Bob,REP}}^2$ and $d_{\text{Bob,SUP}}^2$, the PSA performance gap between the Auth-SUP and Auth-REP schemes increases as $\gamma_{i,\text{Eve}}$ increases.*

For the Auth-SUP scheme, the property mentioned in Corollary 1 is desired, especially when Eve has a better channel quality between Alice and Eve.

We provide an example to show the effect of γ_{Eve} on the PSA of both the Auth-SUP and Auth-REP schemes, as shown in Fig. 11, where $L_R = 6$ and $\rho_t^2 = L_R/L$. In Fig. 11a $\gamma_{\text{Eve}} = 10$ dB and in Fig. 11b $\gamma_{\text{Eve}} = 15$ dB. Fig. 11 shows that the Auth-REP scheme is very sensitive to γ_{Eve} , whereas the Auth-SUP scheme maintains obvious robustness. As γ_{Eve} increases, the PSA performance gap between the Auth-SUP and Auth-REP schemes increases.

Corollary 2. *Because L only affects $d_{\text{Eve,REP}}^2$, the PSA of the Auth-REP scheme increases as L increases. The improvement in the Auth-REP scheme is more obvious than that in the Auth-SUP scheme.*

We provide another example to show the effect of L on the PSA of both the Auth-SUP and Auth-REP schemes, as shown in Fig. 12, where the remaining parameters are the same as those shown in Fig. 11b except for $L = 256$. By comparing the results in Fig. 11b with those in Fig. 12, we can see that the PSA of the Auth-REP scheme increases as L increases, and approaches the PSA of the Auth-SUP scheme. Meanwhile, the Auth-SUP scheme maintains obvious robustness.

In the final example, we consider the case where Eve is a smart and powerful adversary. She can perfectly estimate the location of Bob and make herself approach Bob, i.e.,

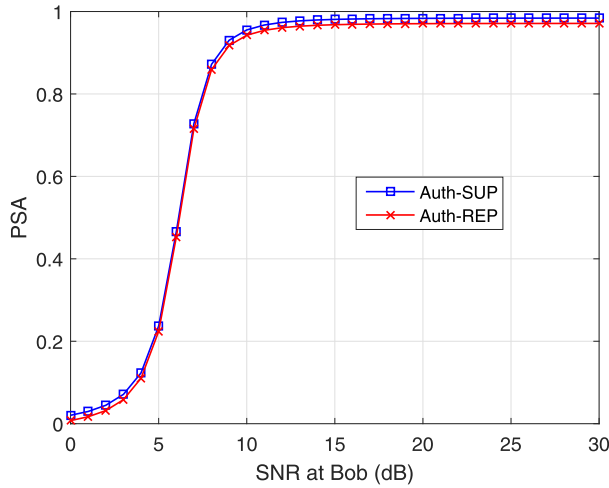


Fig. 12. PSA comparison between the Auth-SUP and Auth-REP schemes versus different Bob's SNRs when the theoretical PFA bound is $\varepsilon_{\text{PFA,Bob}} = \varepsilon_{\text{PFA,Eve}} = 0.01$, where the remaining parameters are the same as those of Fig. 11b except $L = 256$.

$\gamma_{\text{Eve}} = \gamma_{\text{Bob}}$. The results are shown in Fig. 13, where $\gamma_{\text{Eve}} = \gamma_{\text{Bob}}$, $L_R = 6$ and $\rho_t^2 = L_R/L$. Fig. 13 shows that both schemes obviously deteriorate as SNR increases. However, the Auth-SUP scheme still displays better performance than the Auth-REP scheme in the higher SNR regions.

Corollary 3. *The parameter selection of the Auth-SUP scheme is more flexible than that of the Auth-REP scheme, because ρ_s or ρ_t can have an arbitrary value over $[0, 1]$, whereas L_R should be an integer.*

In summary, on the basis of the proposed generic security model, we can draw the following conclusion. In comparison with the Auth-REP scheme, the Auth-SUP scheme is a more powerful, robust and flexible solution.

7 CONCLUSION

This paper has presented a new generic security model of PHY-layer authentication by considering the covertness, security and robustness together. On the basis of the proposed security model, a new systematic metric is designed, which is referred to as the PSA. Based on the proposed generic security model, we can not only systematically analyze the effect of the parameter of a certain PHY-layer authentication scheme on the final performance, but also fairly compare the performance of different PHY-layer authentication schemes under the same channel conditions at the legitimate and adversary receivers.

In addition to the Auth-SUP scheme in the literature, we have also given a modified PHY-layer authentication scheme, which is referred to as the Auth-REP scheme, for comparison purposes. For the performance analysis, the PD and PFA of both schemes over a fading channel have been defined, and their closed-form expressions have been derived. From the results of performance comparison by using the proposed generic security model, the Auth-SUP scheme is a better option than the Auth-REP scheme. Optimizing the parameters of certain PHY-layer authentication scheme under the proposed security model is an interesting topic, which would be our future work.

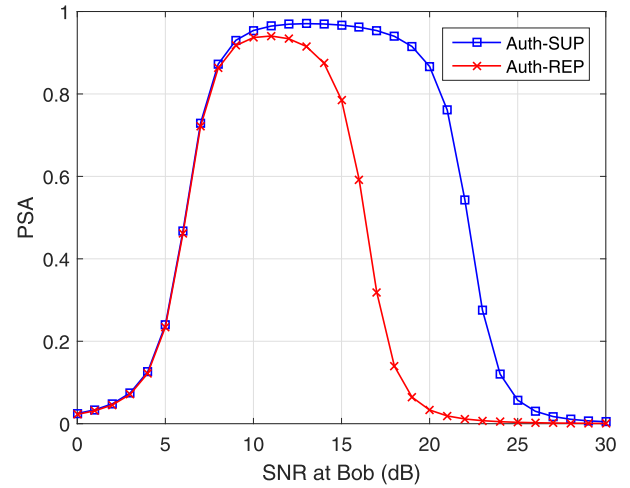


Fig. 13. PSA comparison between the Auth-SUP and Auth-REP schemes versus different Bob's SNRs when the theoretical PFA bound is $\varepsilon_{\text{PFA,Bob}} = \varepsilon_{\text{PFA,Eve}} = 0.01$, where $\gamma_{\text{Eve}} = \gamma_{\text{Bob}}$, $L_R = 6$ and $\rho_t^2 = L_R/L$.

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